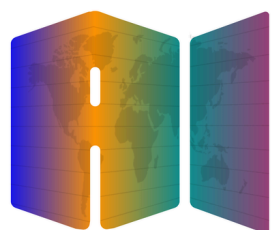




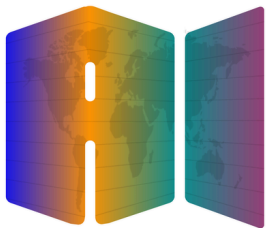
Agentic AI

Lectures by JADVCastro



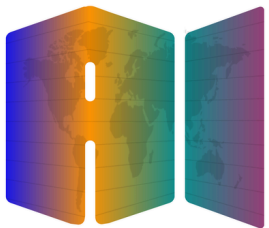
Workflow-as-a-tool Design Pattern

**Using deterministic workflows
as LLM tools**

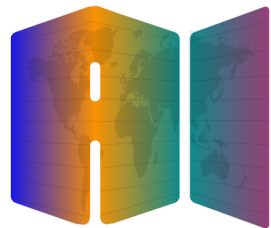
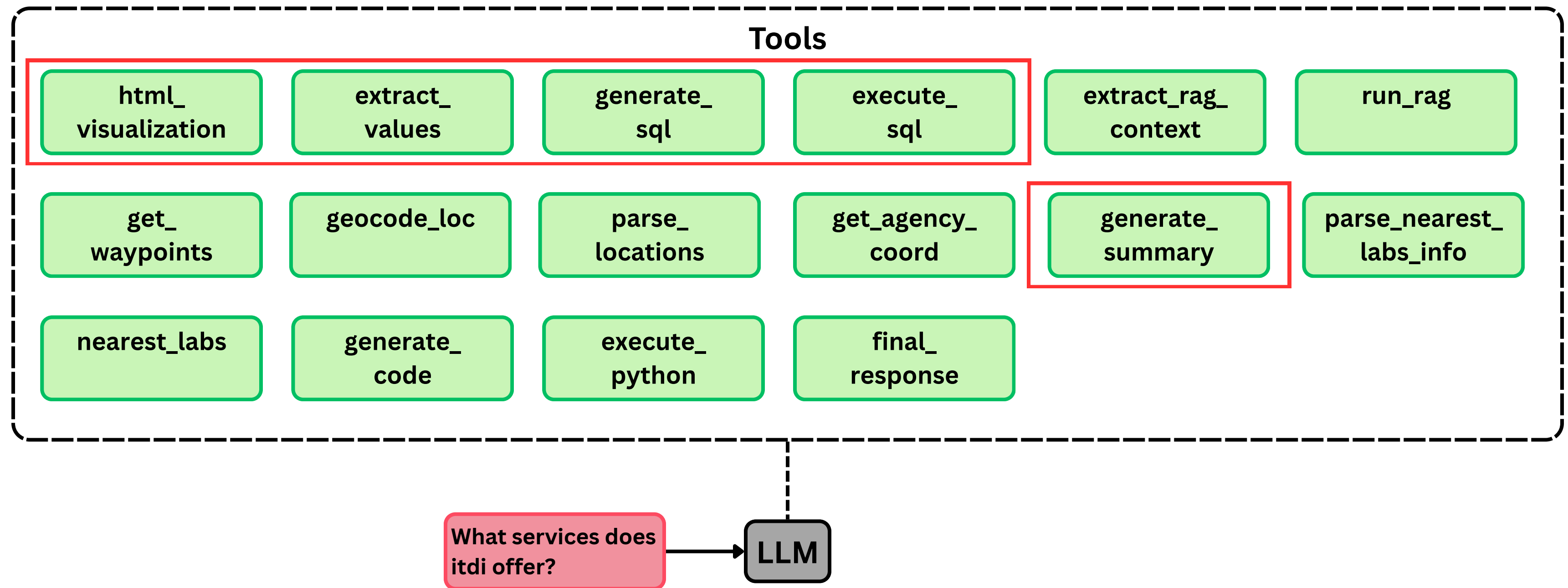


Using deterministic workflows as LLM tools

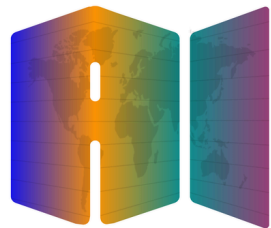
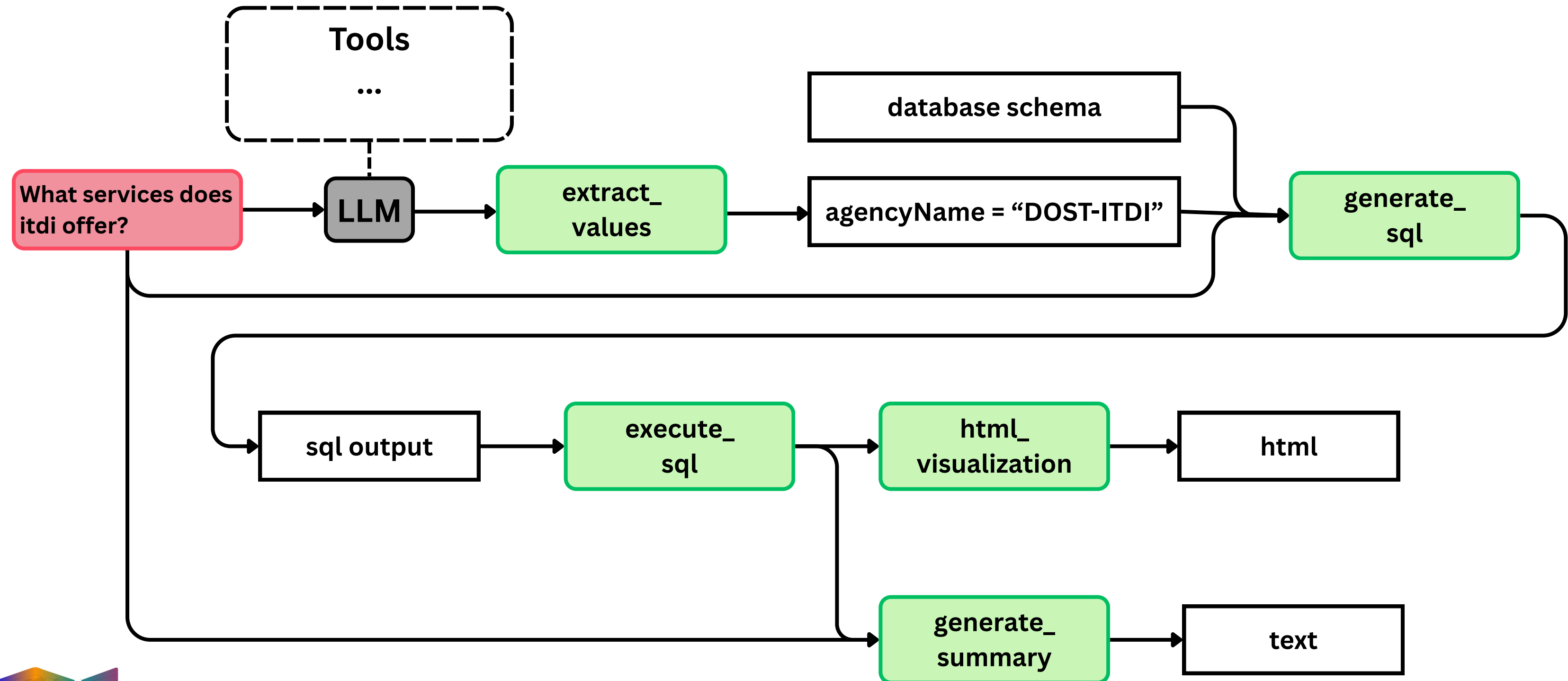
The core idea is that instead of asking the LLM to reason through every step from scratch, you encapsulate a known sequence of actions into a single callable tool



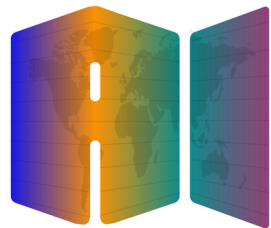
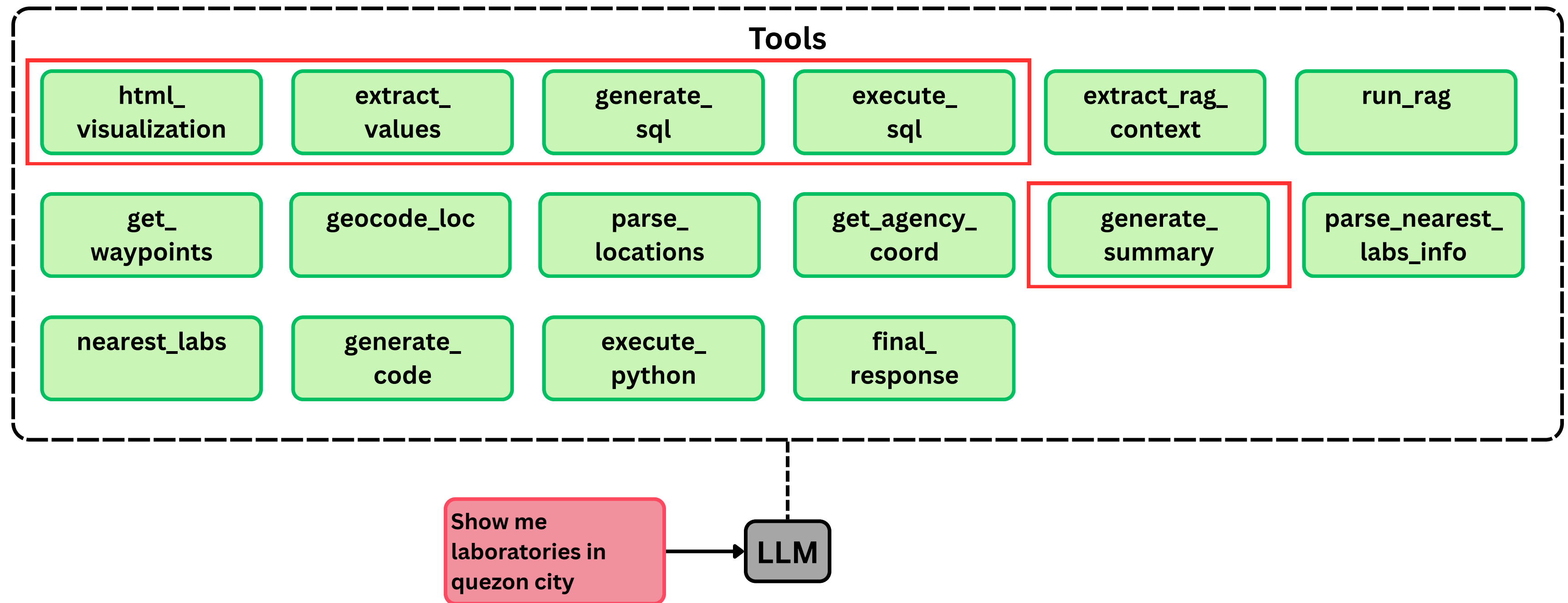
Using deterministic workflows as LLM tools



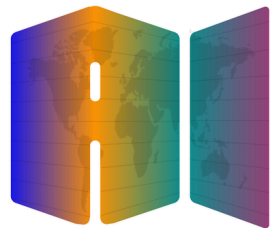
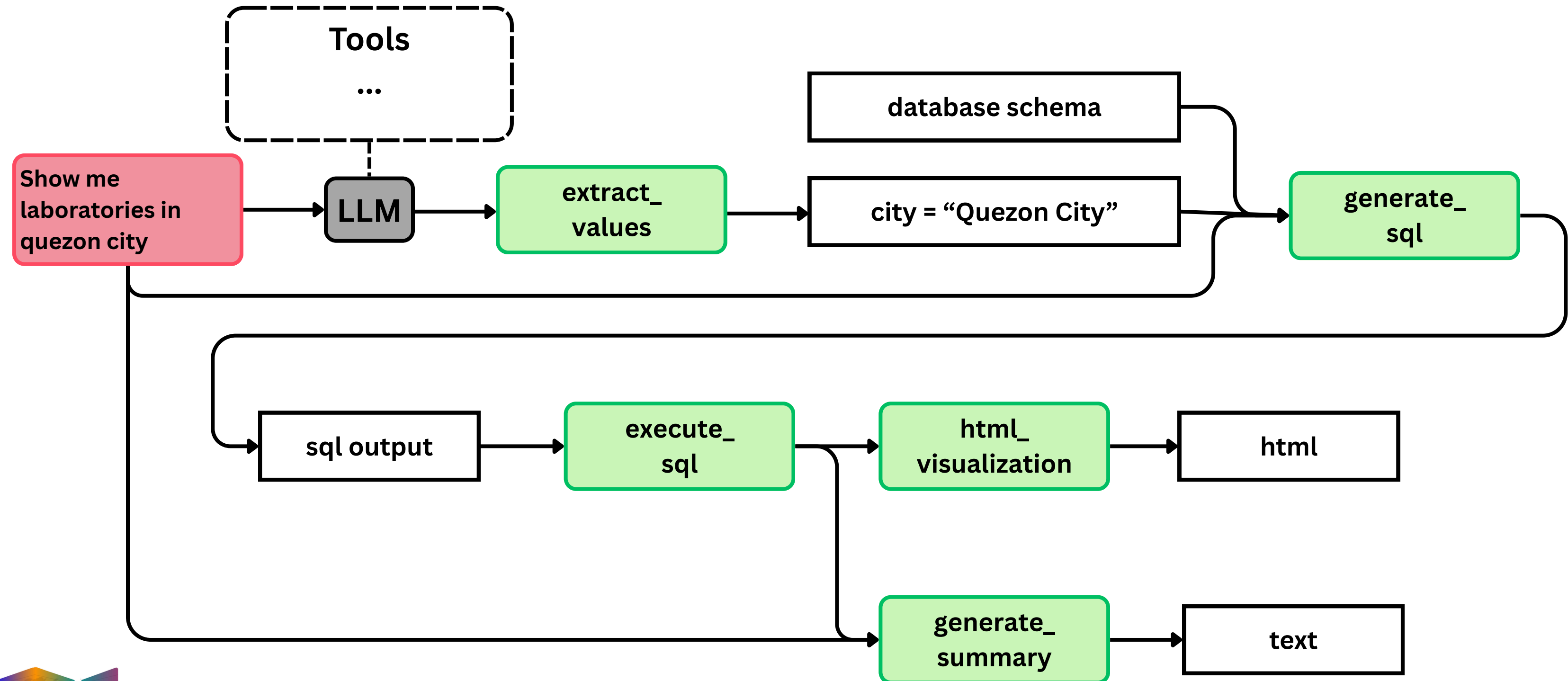
Using deterministic workflows as LLM tools



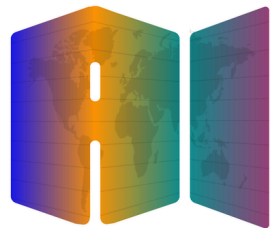
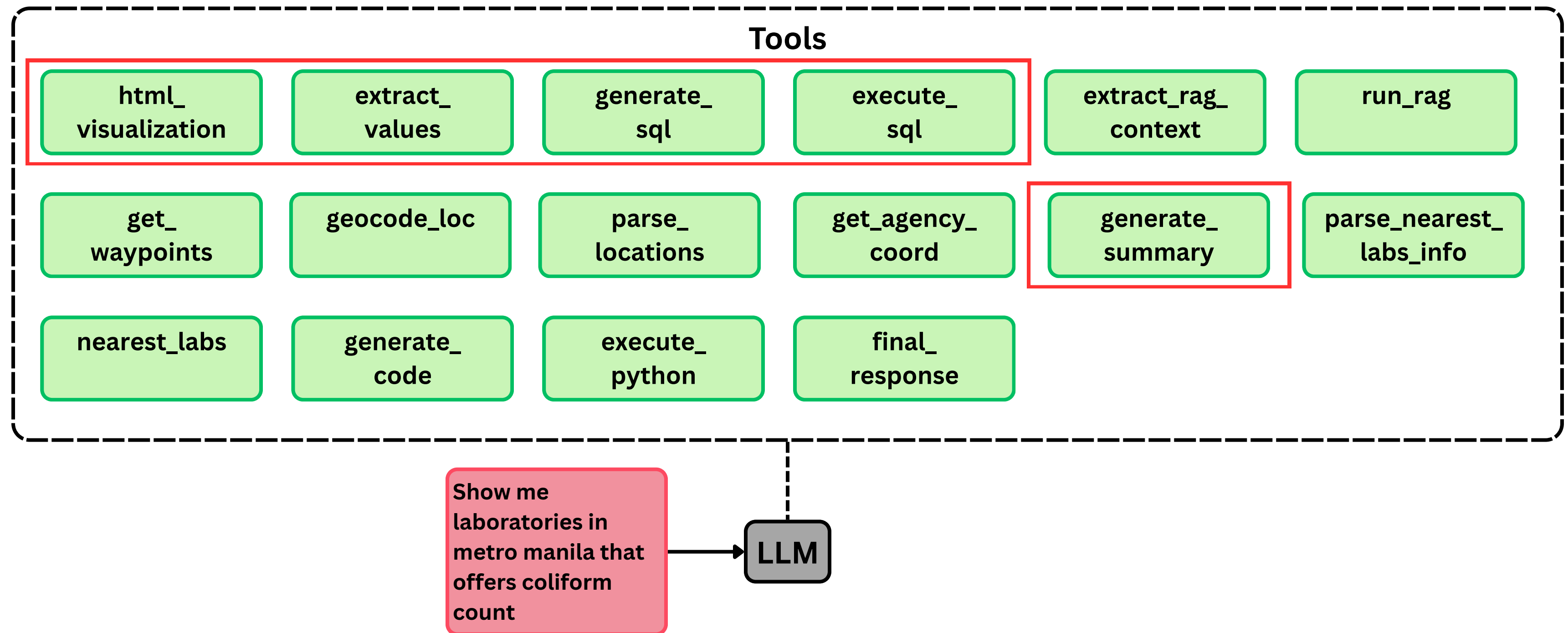
Using deterministic workflows as LLM tools



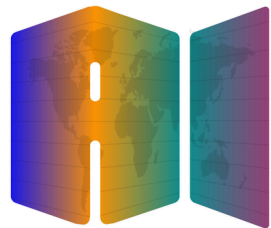
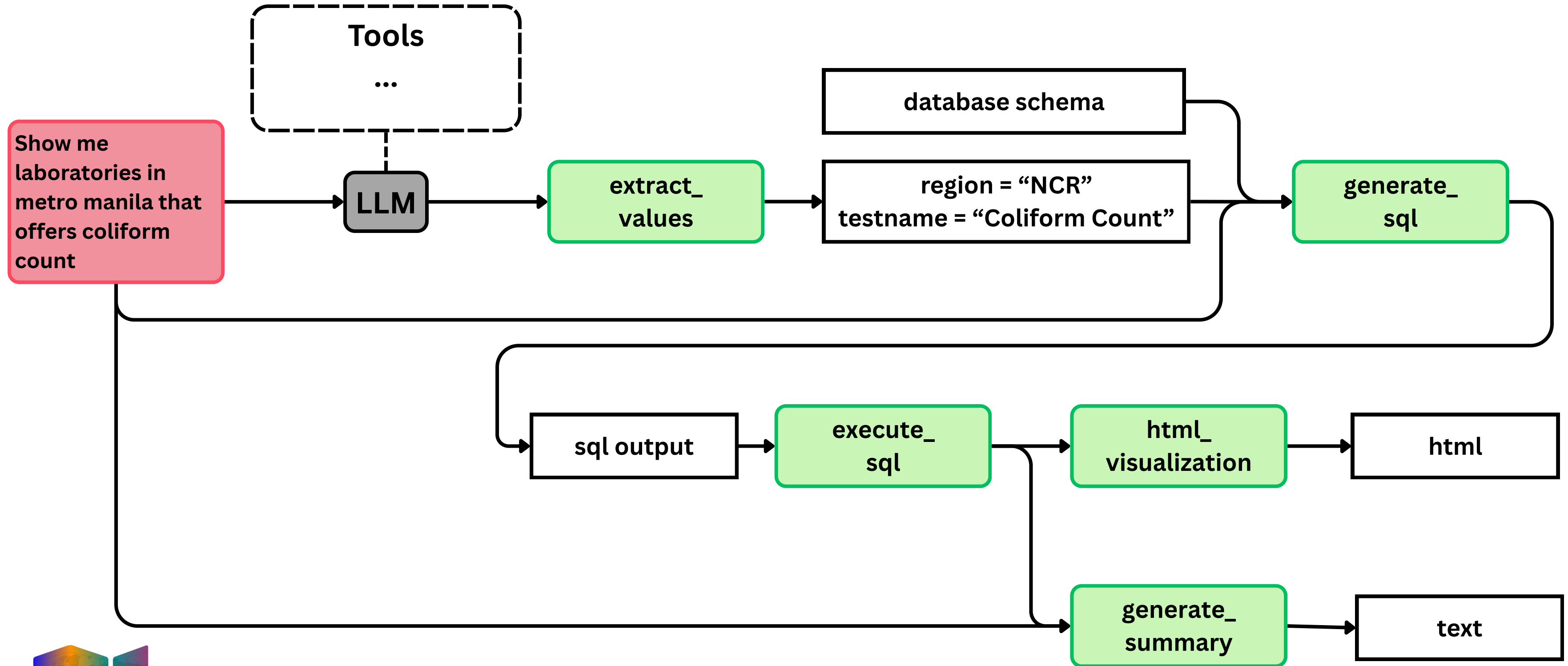
Using deterministic workflows as LLM tools



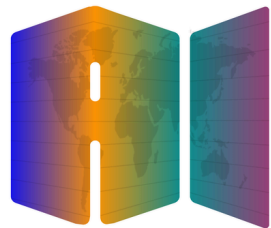
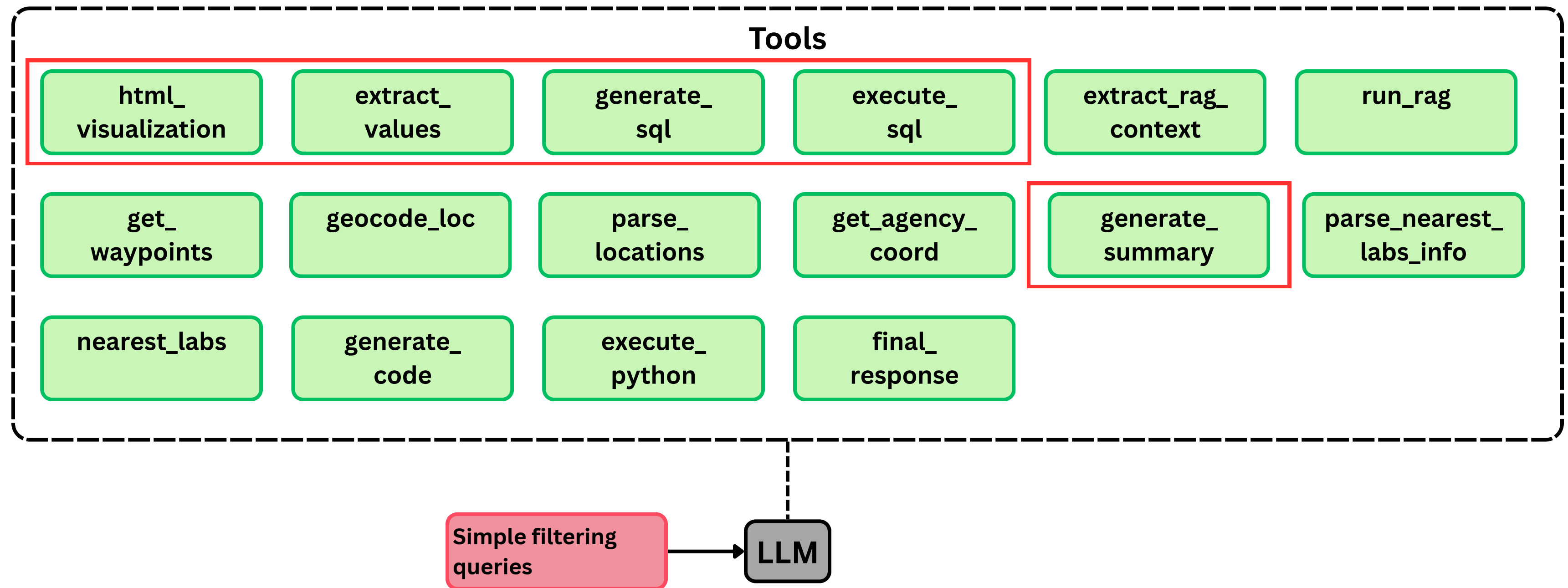
Using deterministic workflows as LLM tools



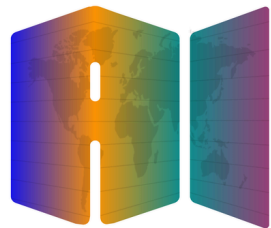
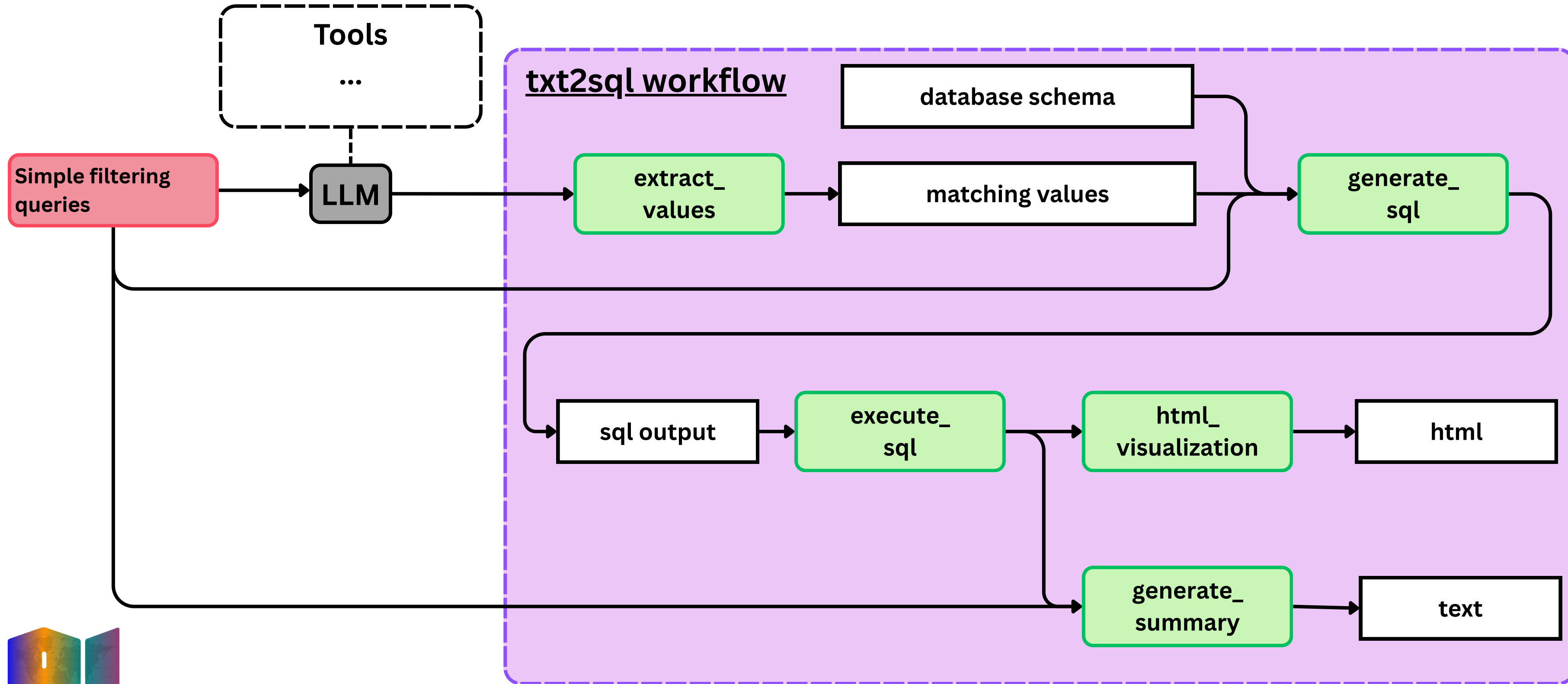
Using deterministic workflows as LLM tools



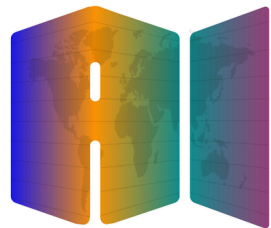
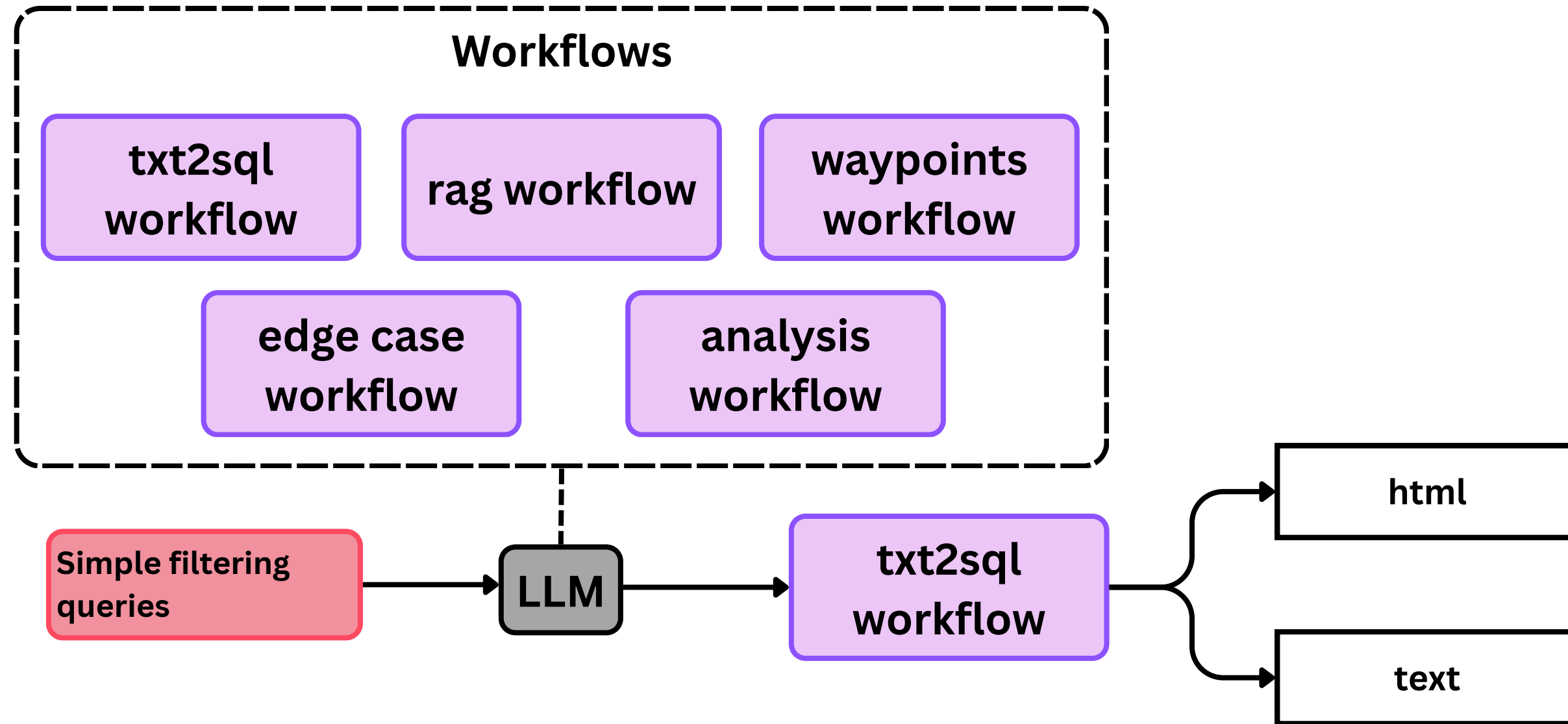
Using deterministic workflows as LLM tools



Using deterministic workflows as LLM tools



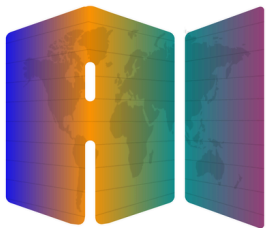
Using deterministic workflows as LLM tools



The LLM performs intent determination and enforces guardrails

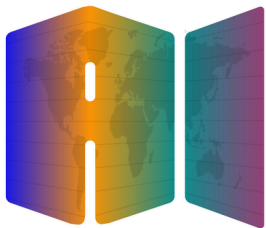
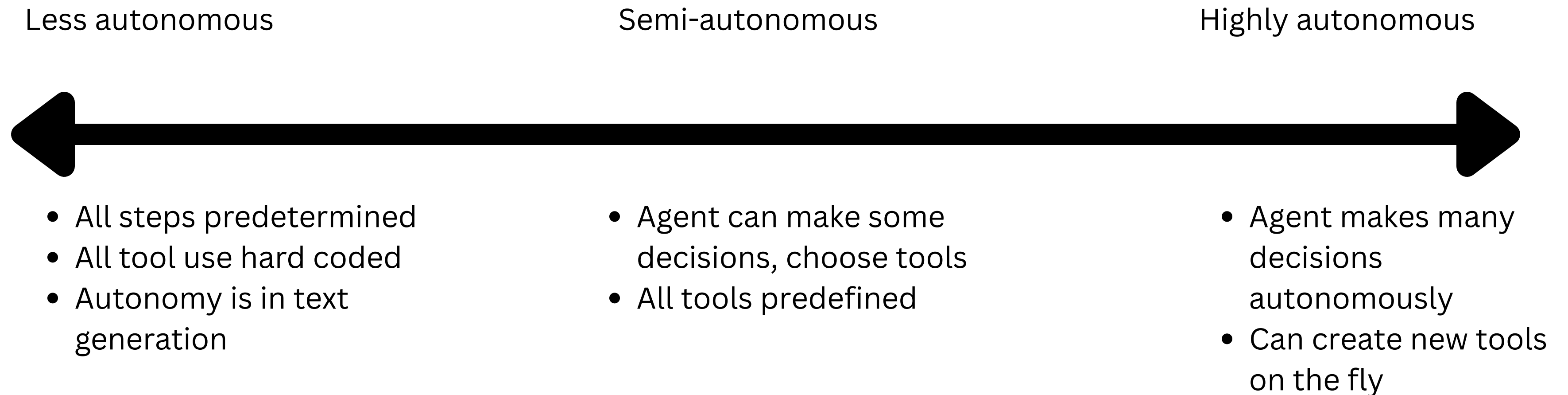
Using deterministic workflows as LLM tools

How is Workflow-as-a-tool different from Agent-as-a-tool?



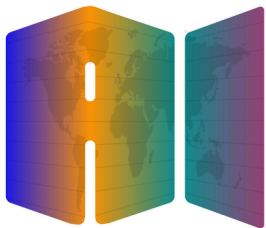
Degrees of autonomy

Agentic AI can be less or more autonomous



Workflow-as-a-tool Design Pattern

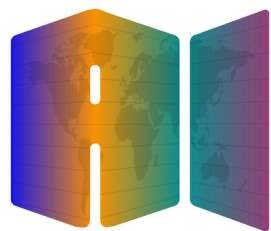
**Advantages of using
deterministic workflows**



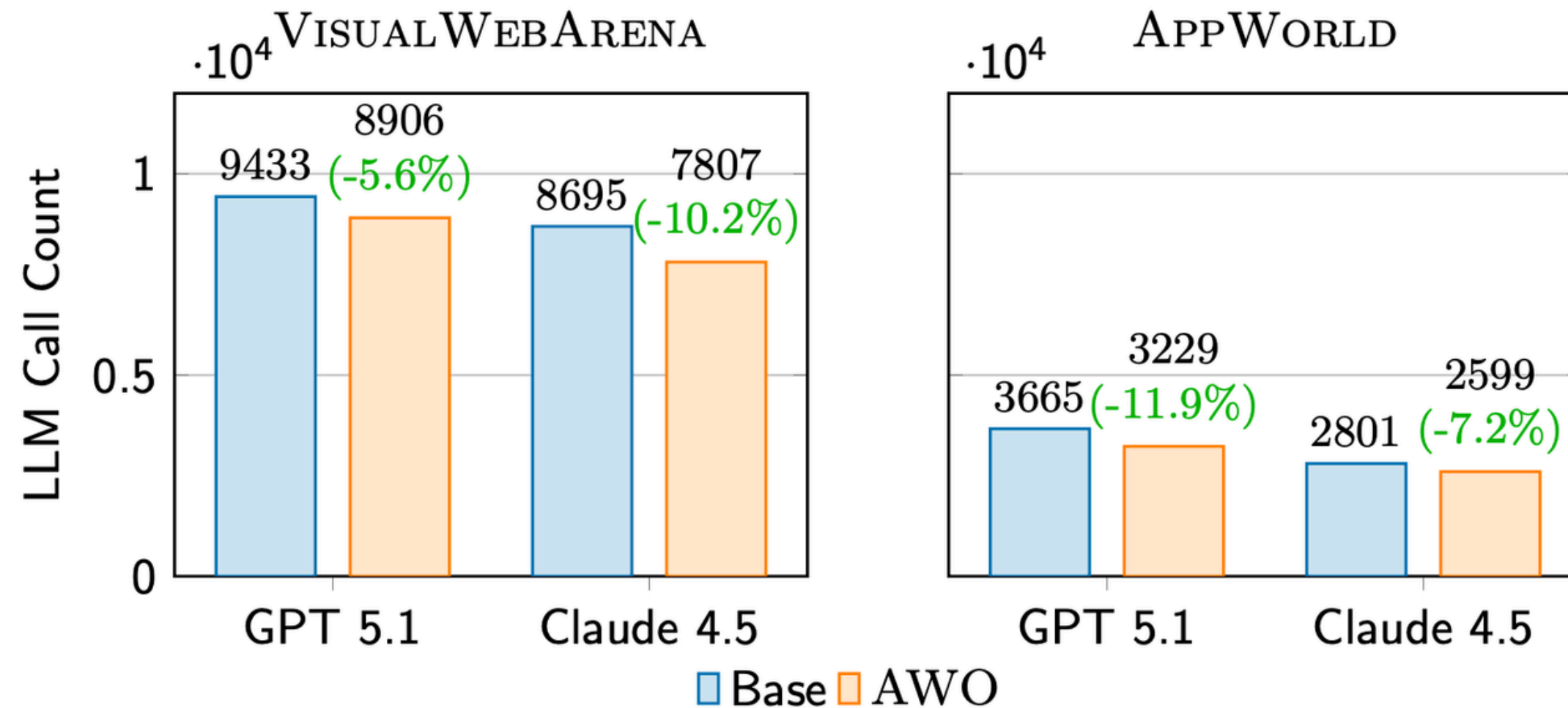
Advantages of using deterministic workflows

Benchmark	Metric	GPT 5.1		Claude 4.5	
		Base	with AWO	Base	with AWO
VisualWebArena	Total Tokens (M)	95.7	90.3 (-5.6%)	120.2	107.9 (-10.2%)
	Total Cost (\$)	44.5	42.0 (-5.7%)	272.3	244.5 (-10.2%)
AppWorld	Total Tokens (M)	34.1	29.0 (-14.9%)	29.8	27.0 (-9.4%)
	Total Cost (\$)	31.59	26.85 (-15.0%)	29.63	28.39 (-4.2%)

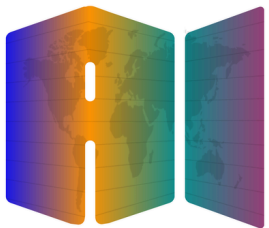
[Adapted from Abuzakuk, S. et al. (2026) “Optimizing Agentic Workflows using Meta-tools”]



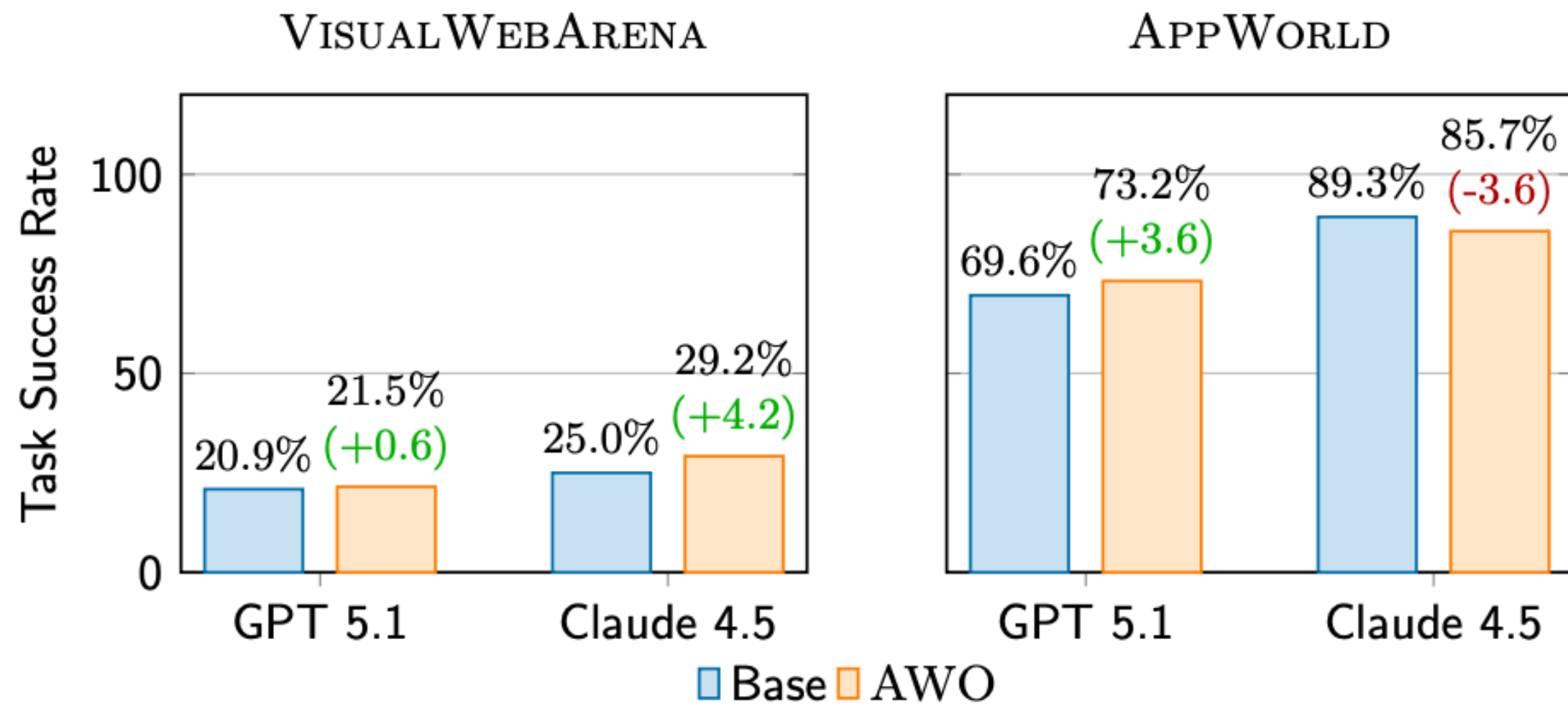
Advantages of using deterministic workflows



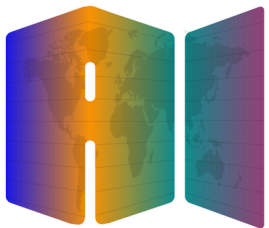
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Advantages of using deterministic workflows

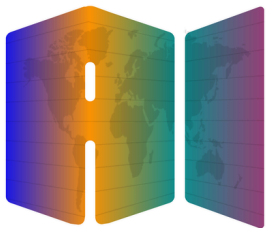


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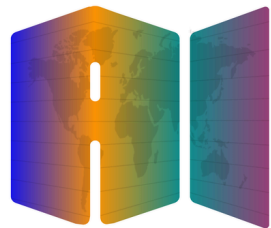
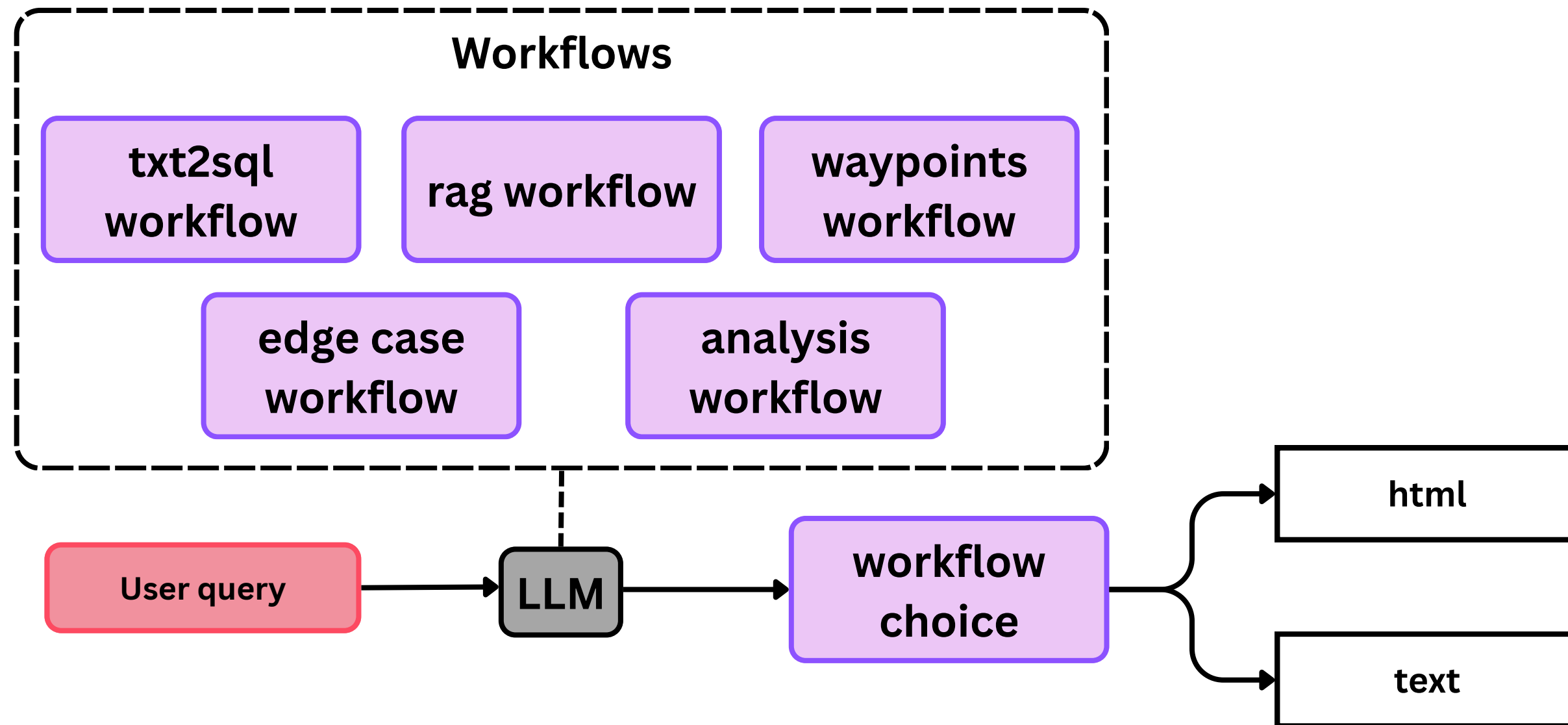


Workflow-as-a-tool Design Pattern

OneLab Agent

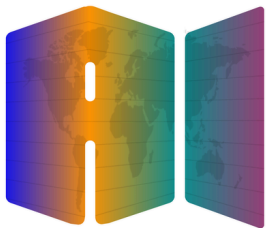


OneLab Agent

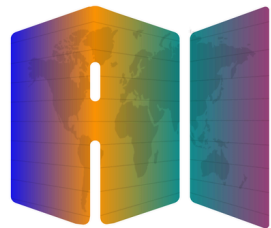
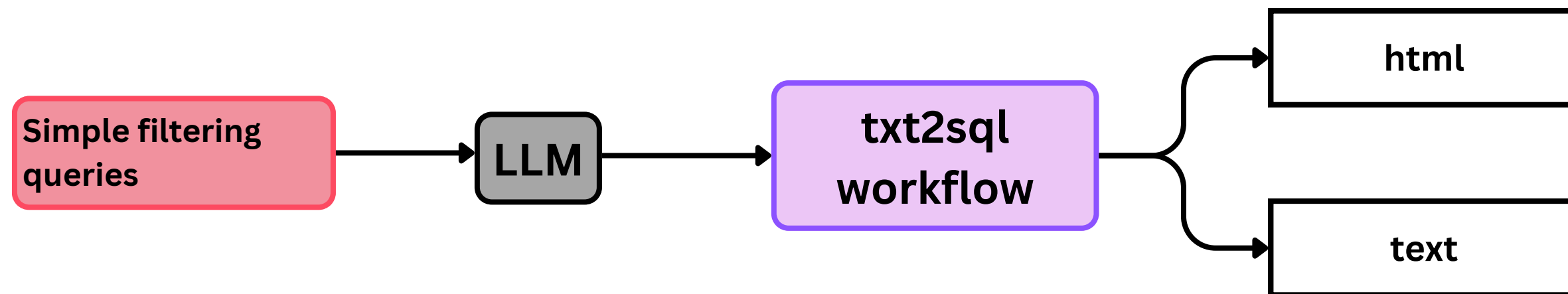


Workflow-as-a-tool Design Pattern

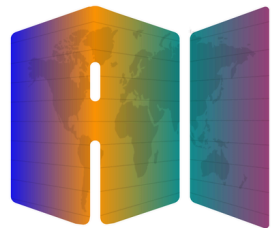
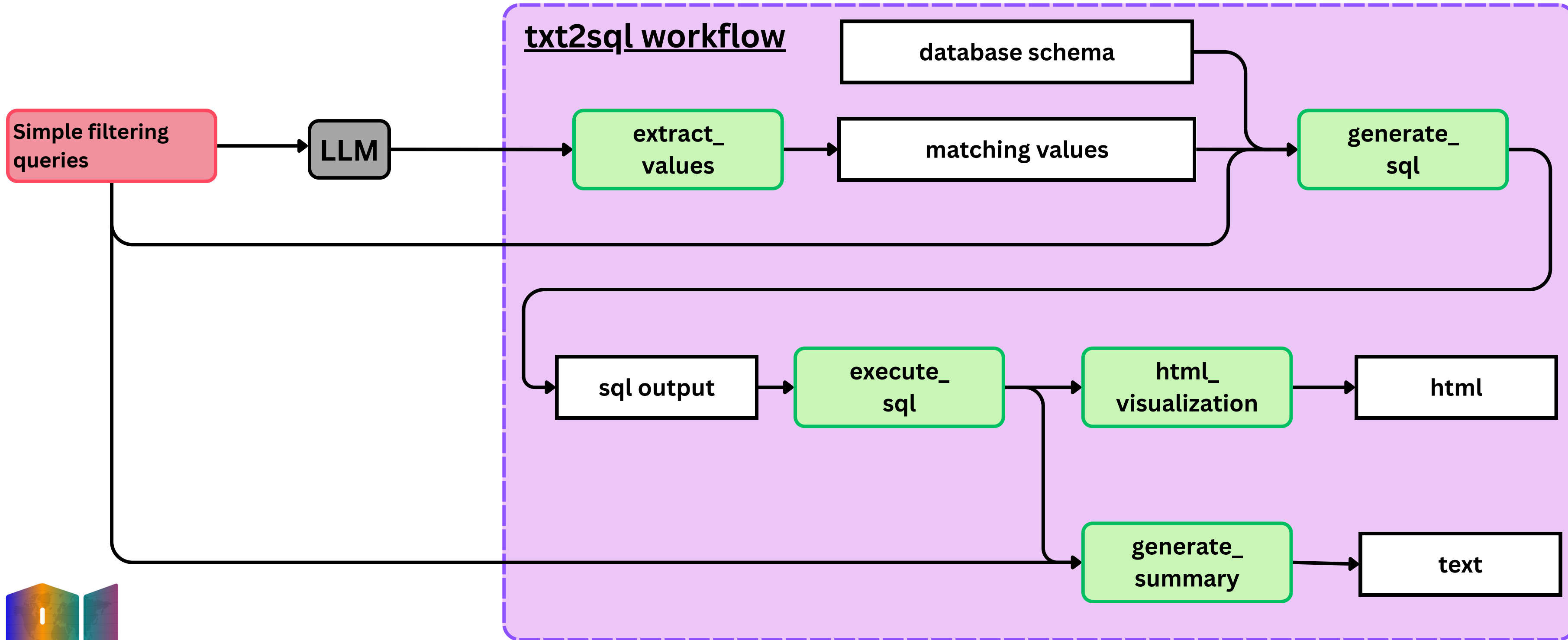
Txt2SQL Workflow



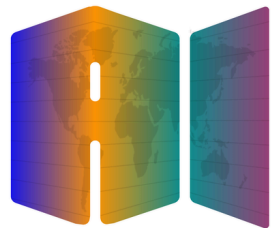
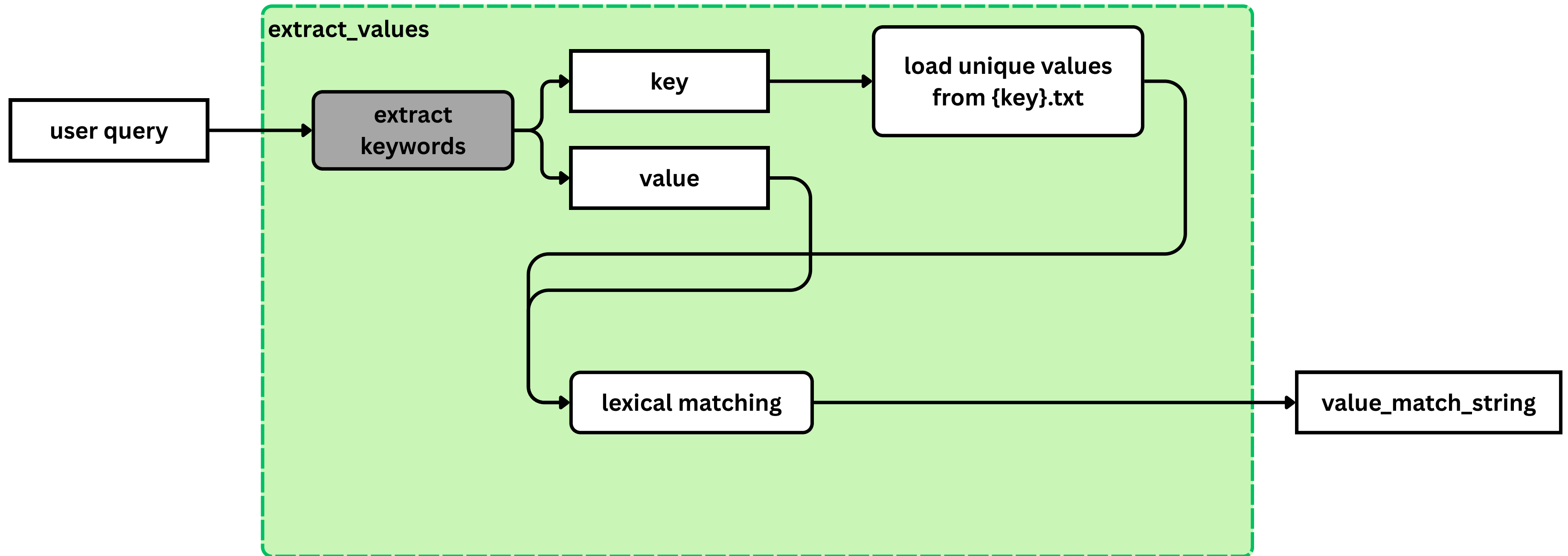
Txt2SQL Workflow



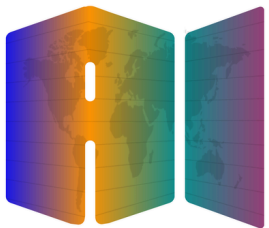
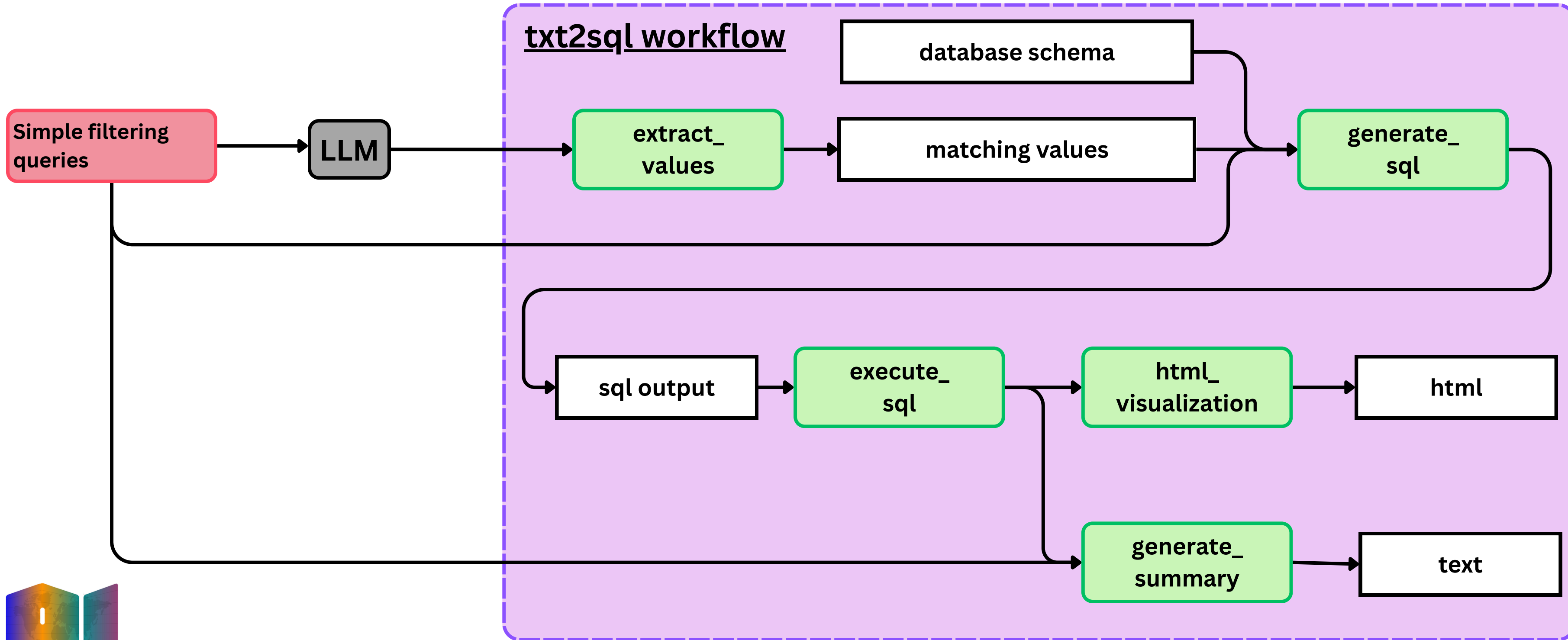
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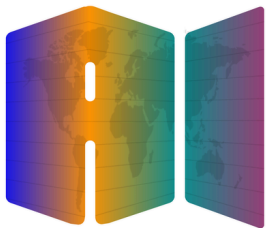
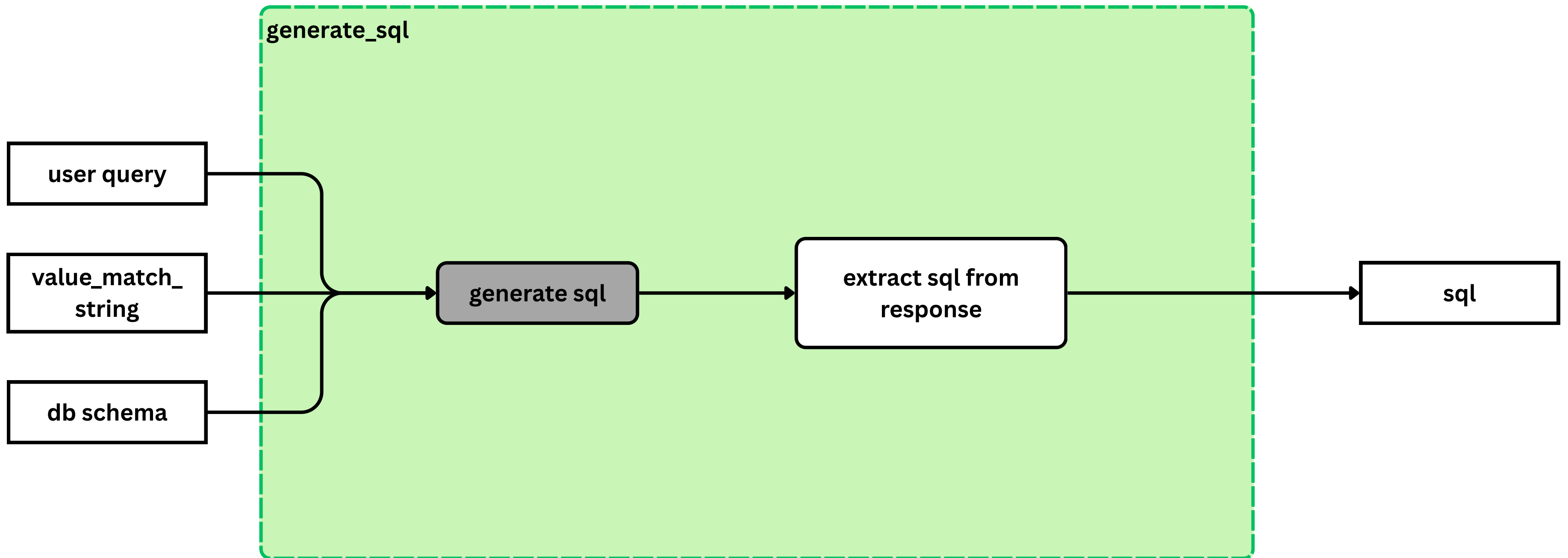
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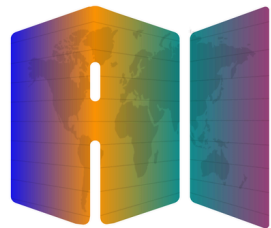
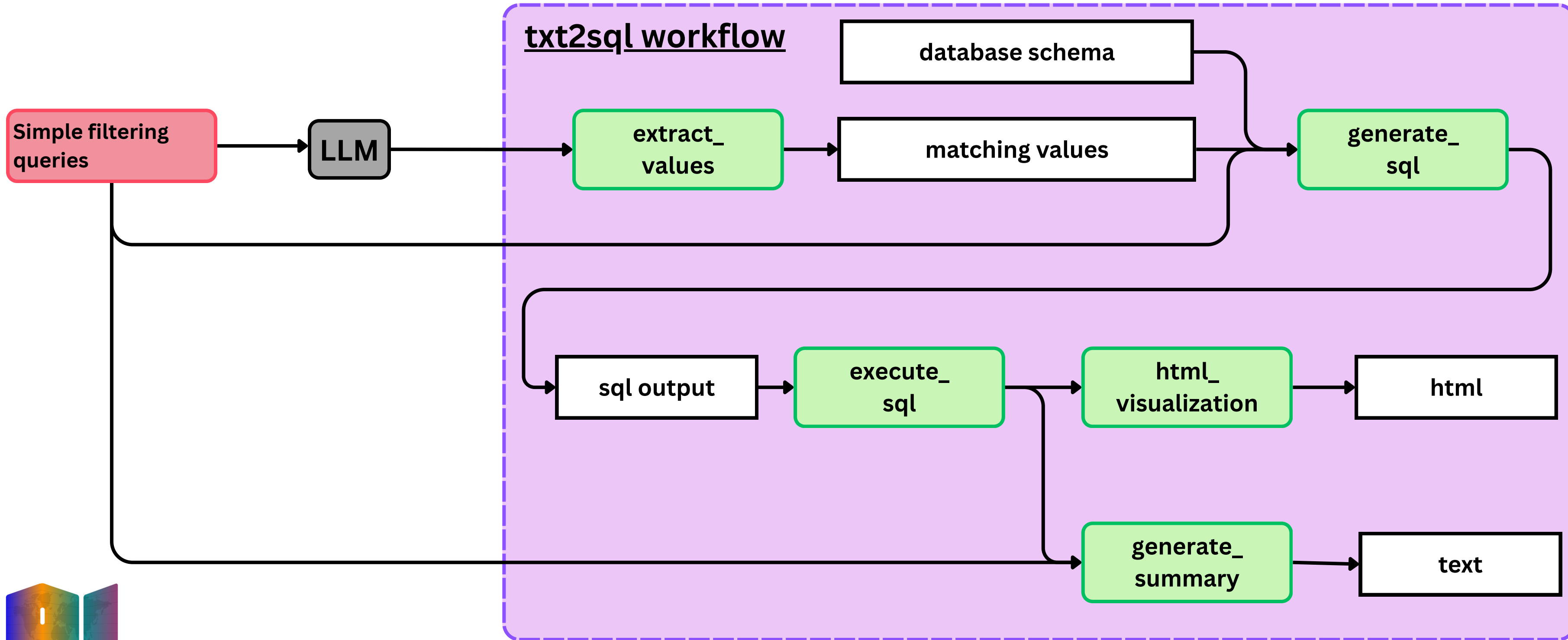
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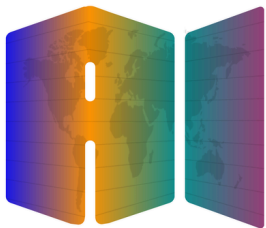
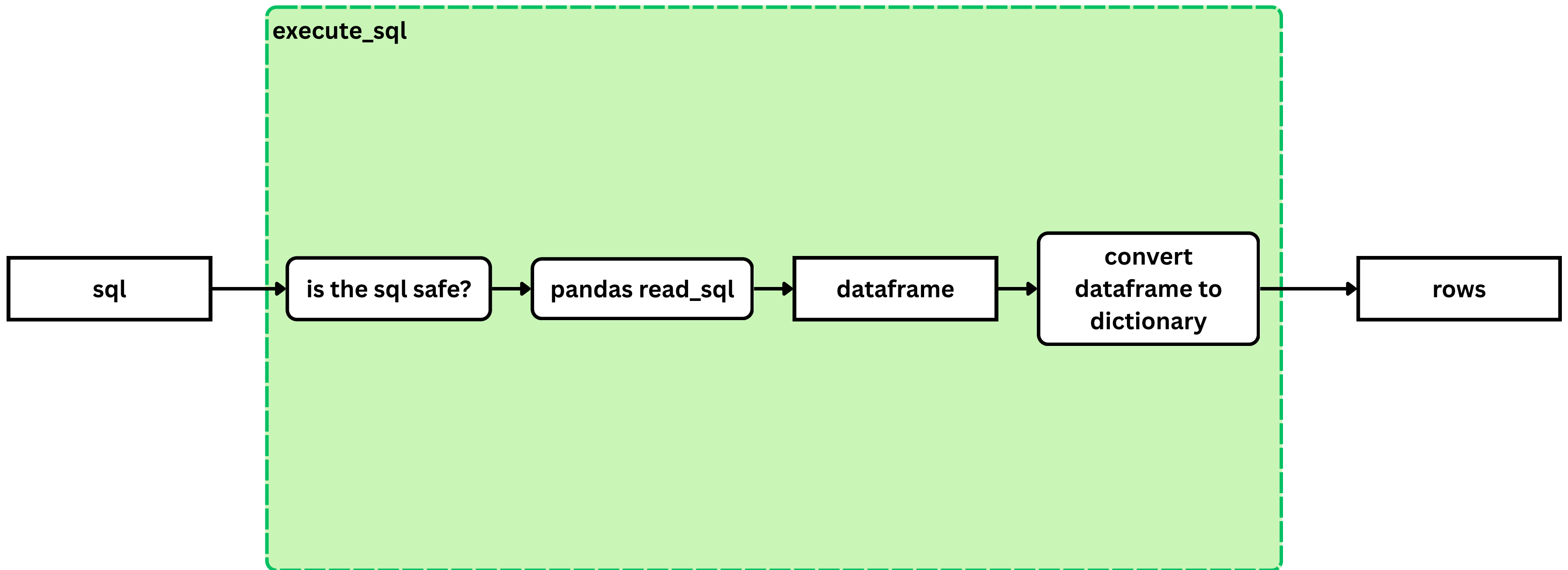
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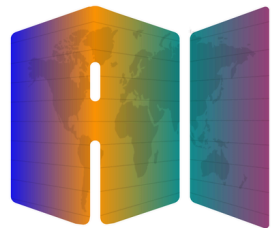
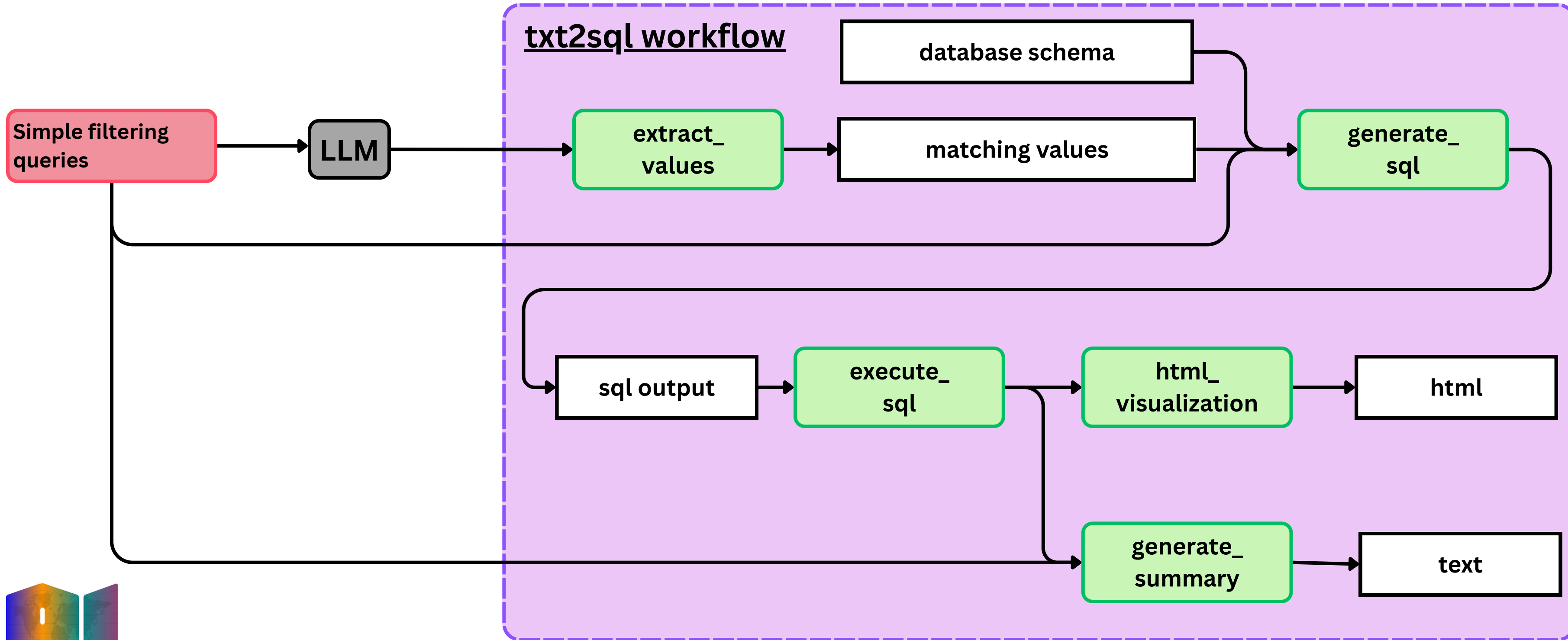
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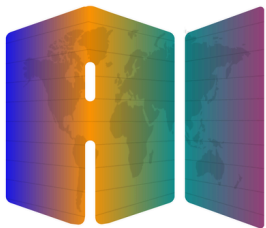
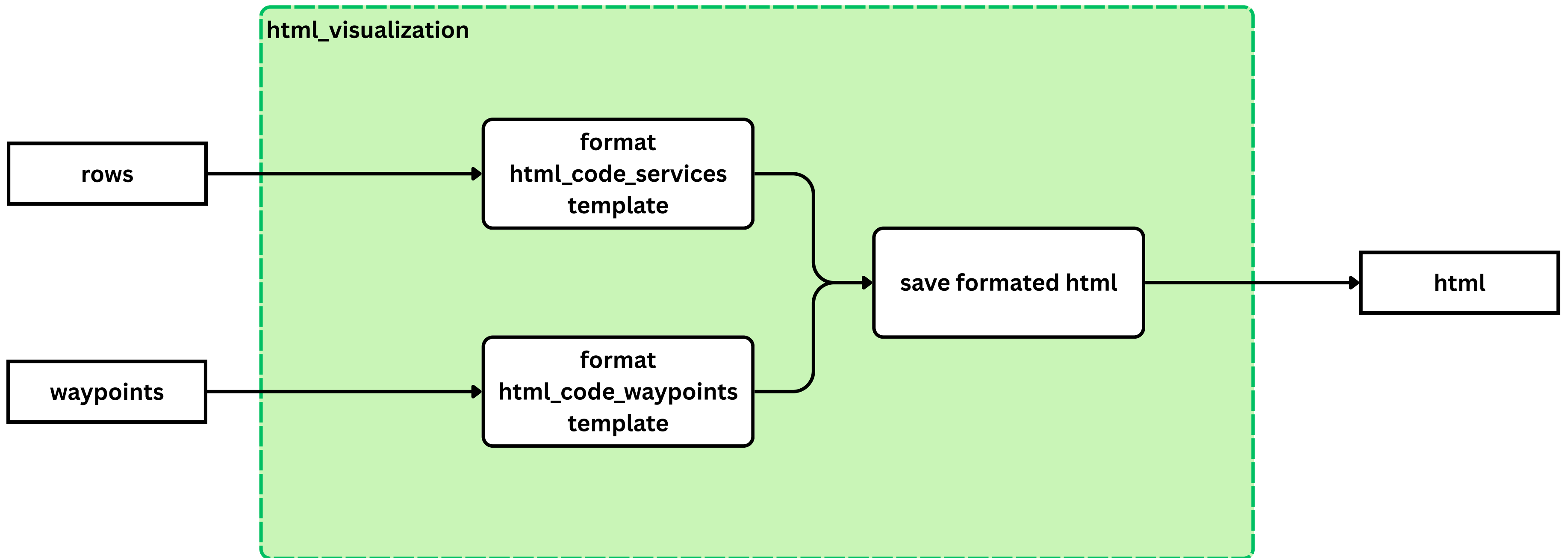
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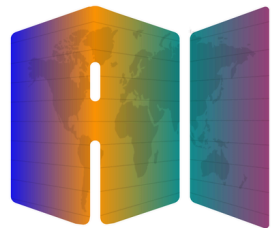
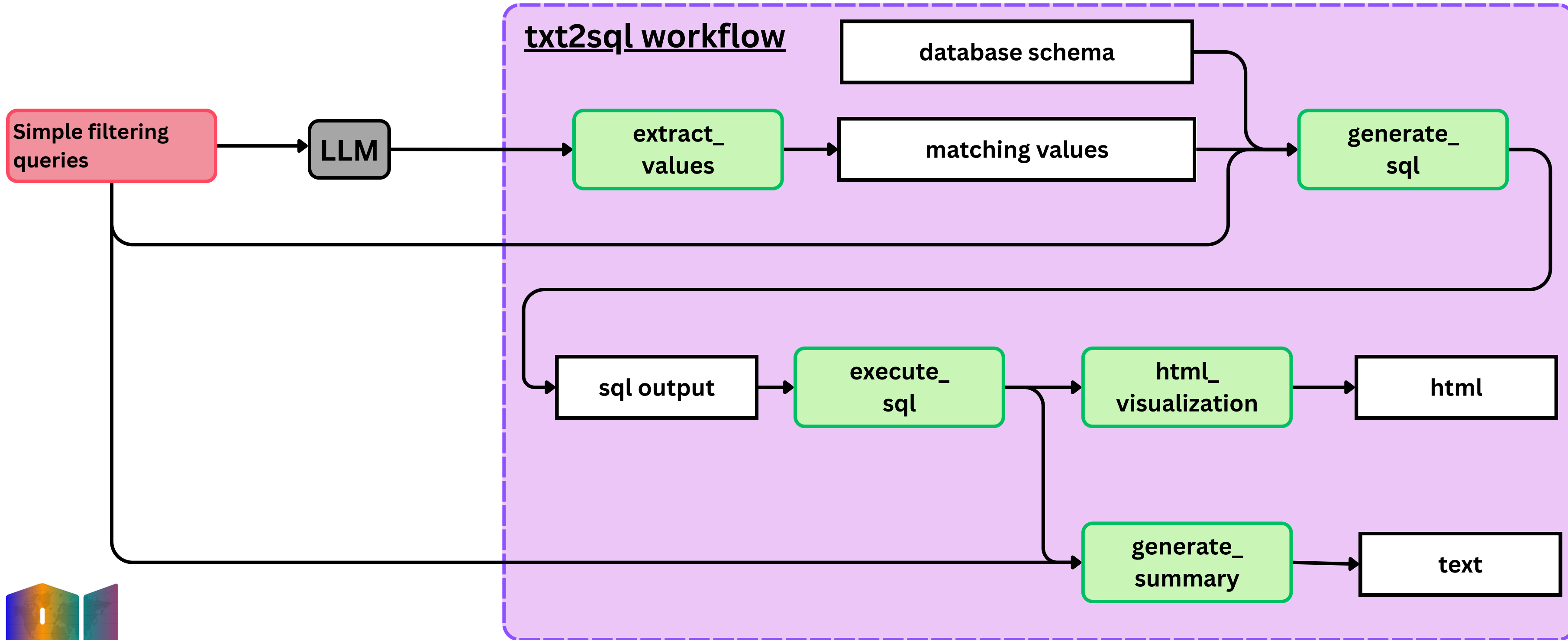
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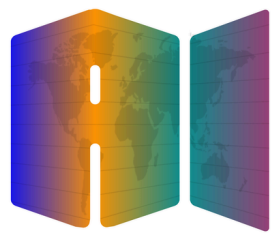
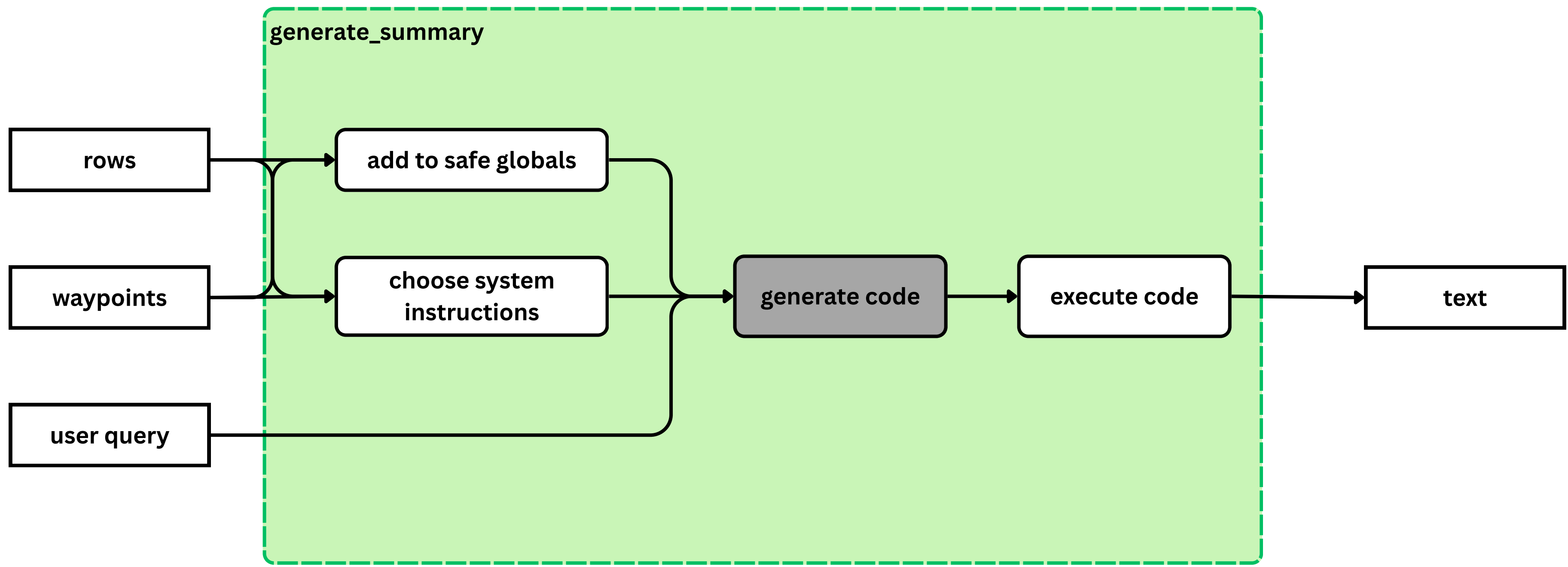
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Txt2SQL Workflow

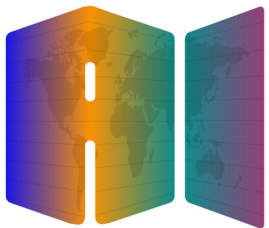


Txt2SQL Workflow

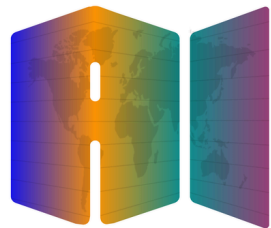
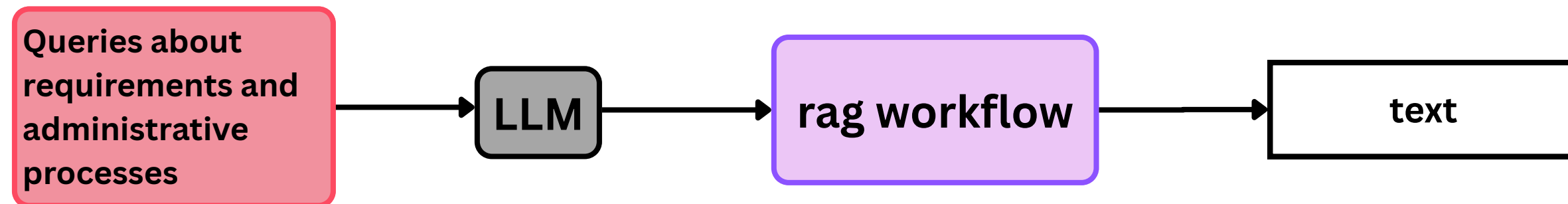


Workflow-as-a-tool Design Pattern

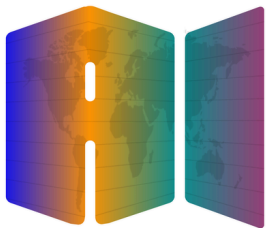
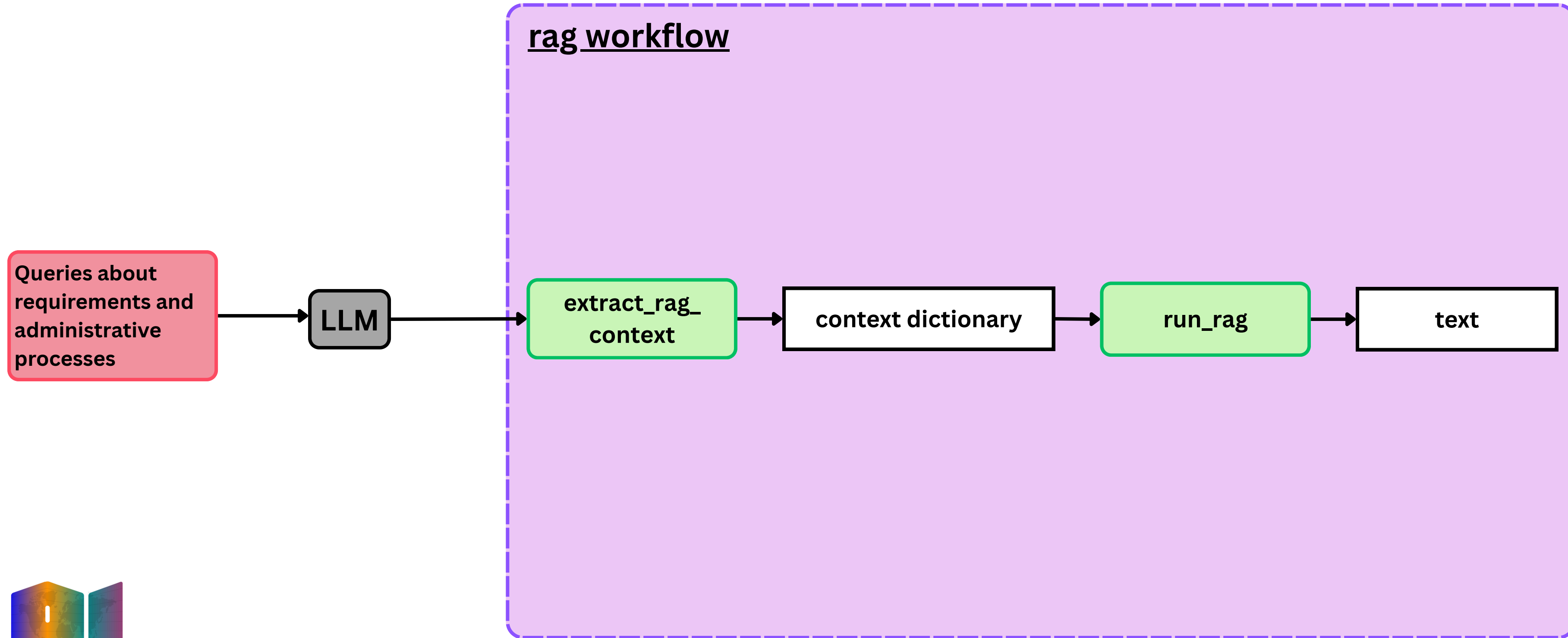
Structured RAG Workflow



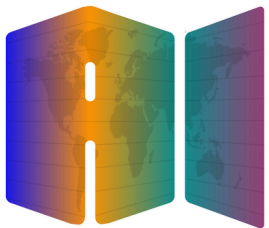
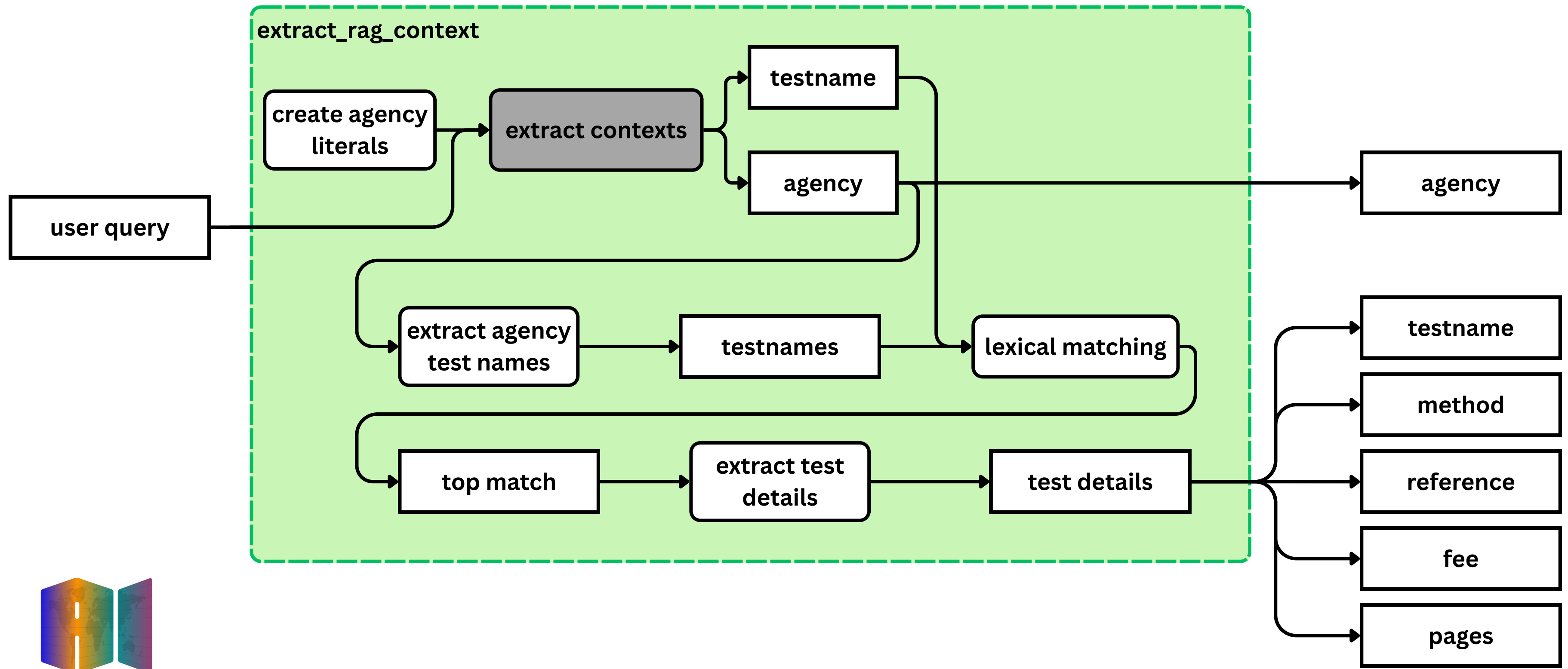
Structured RAG Workflow



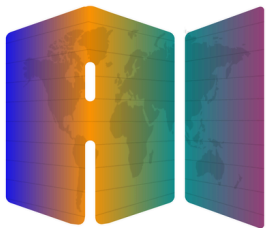
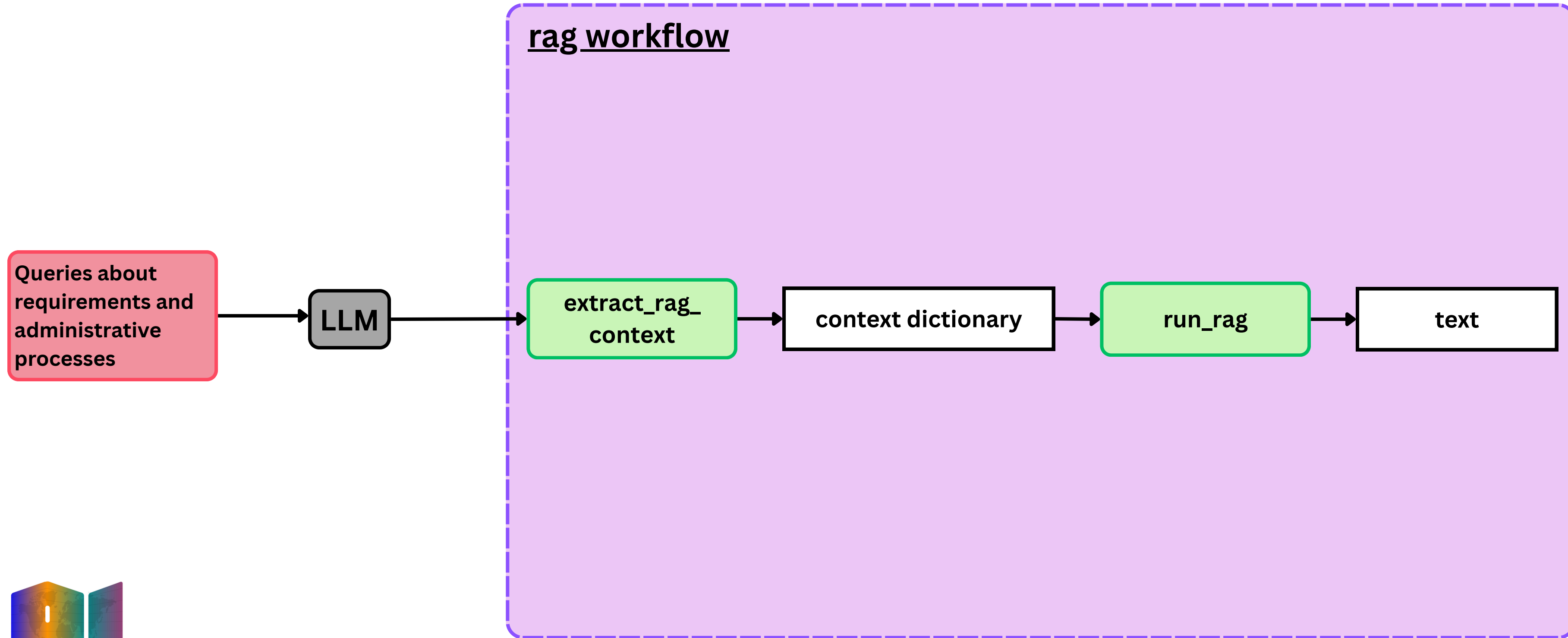
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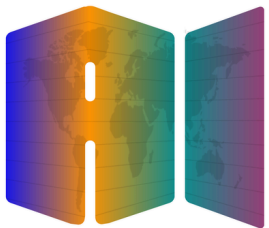
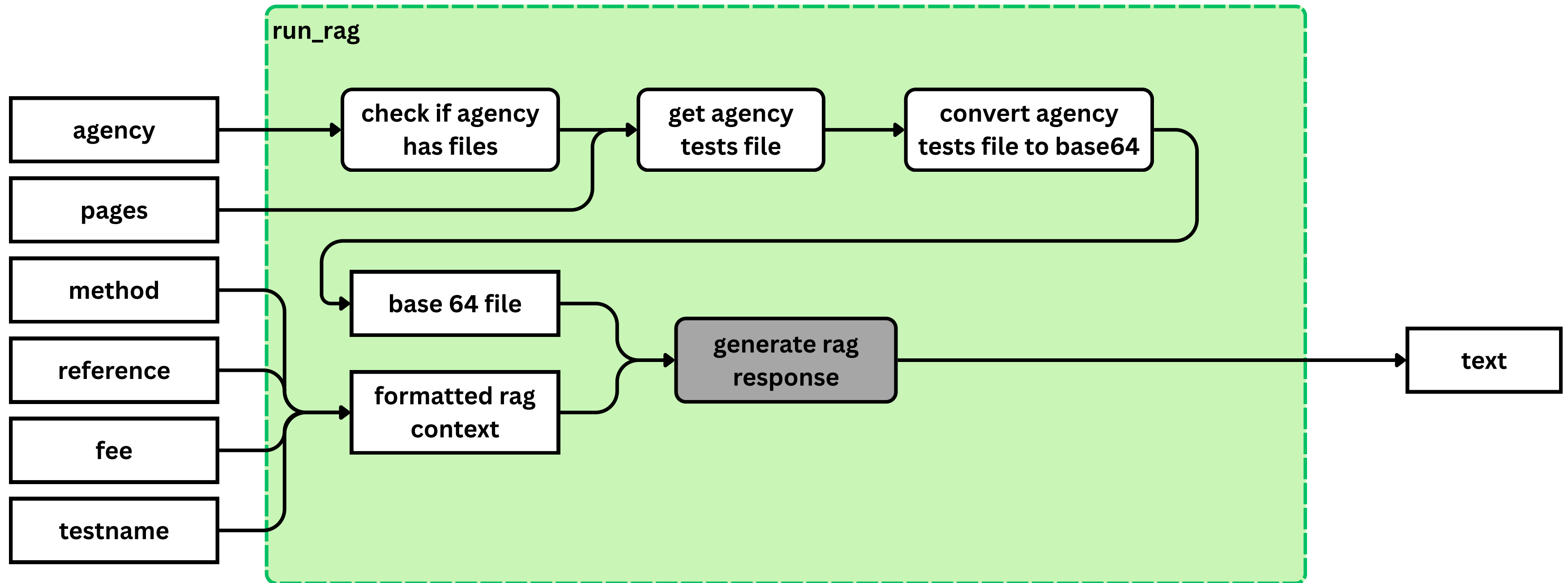
Structured RAG Workflow



Structured RAG Workflow

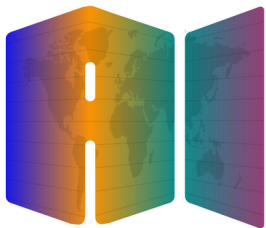


Structured RAG Workflow

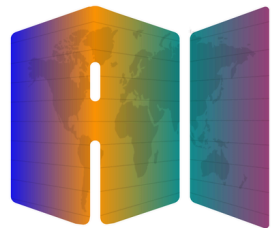
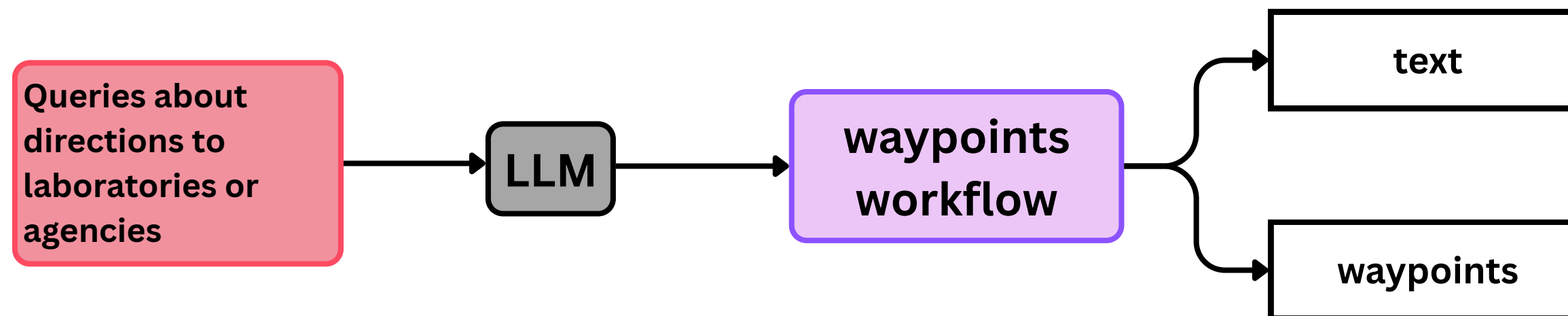


Workflow-as-a-tool Design Pattern

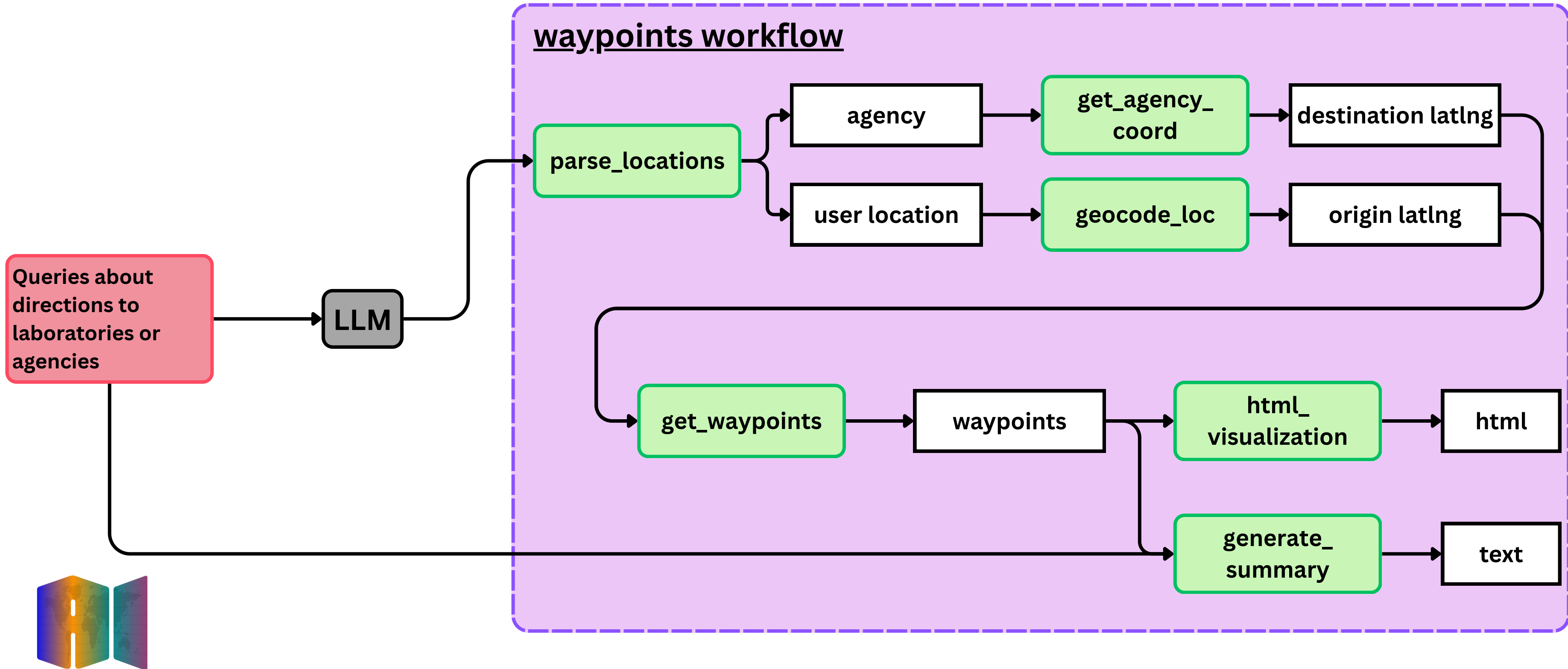
External API Workflow



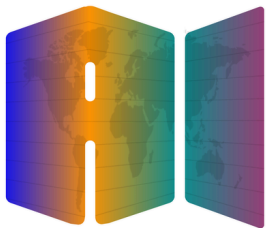
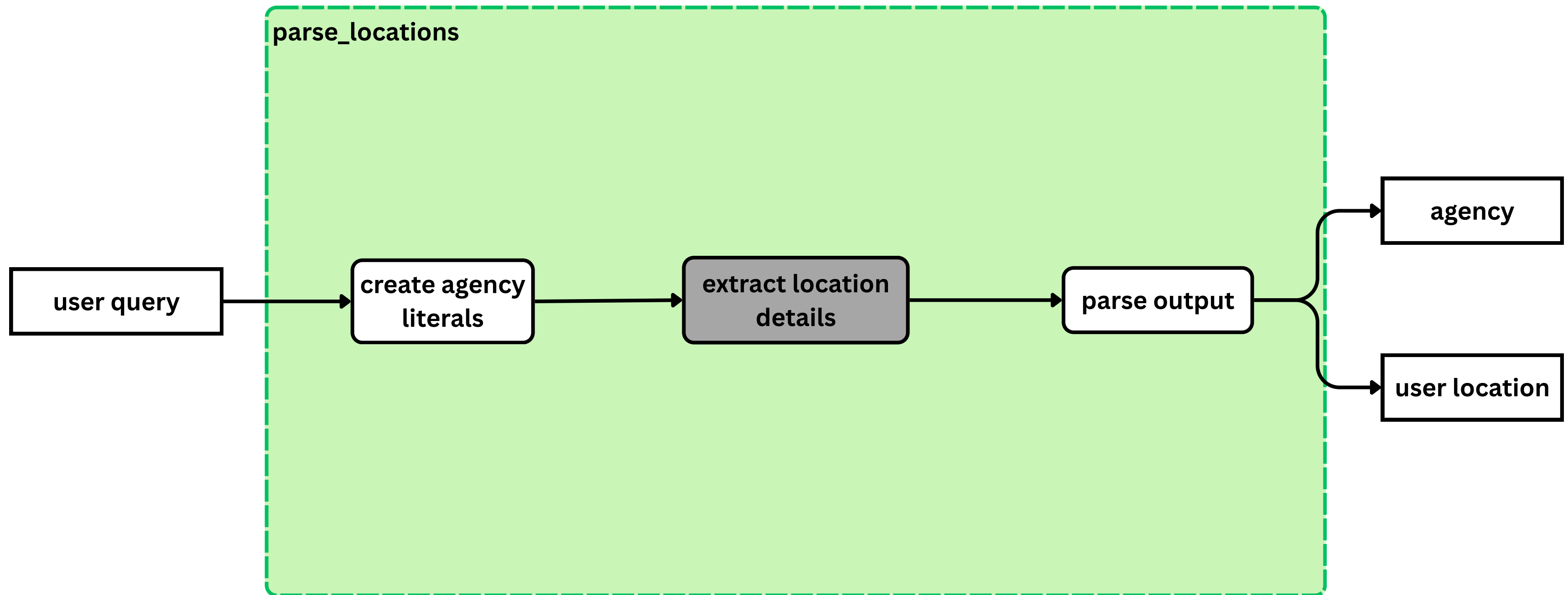
External API Workflow



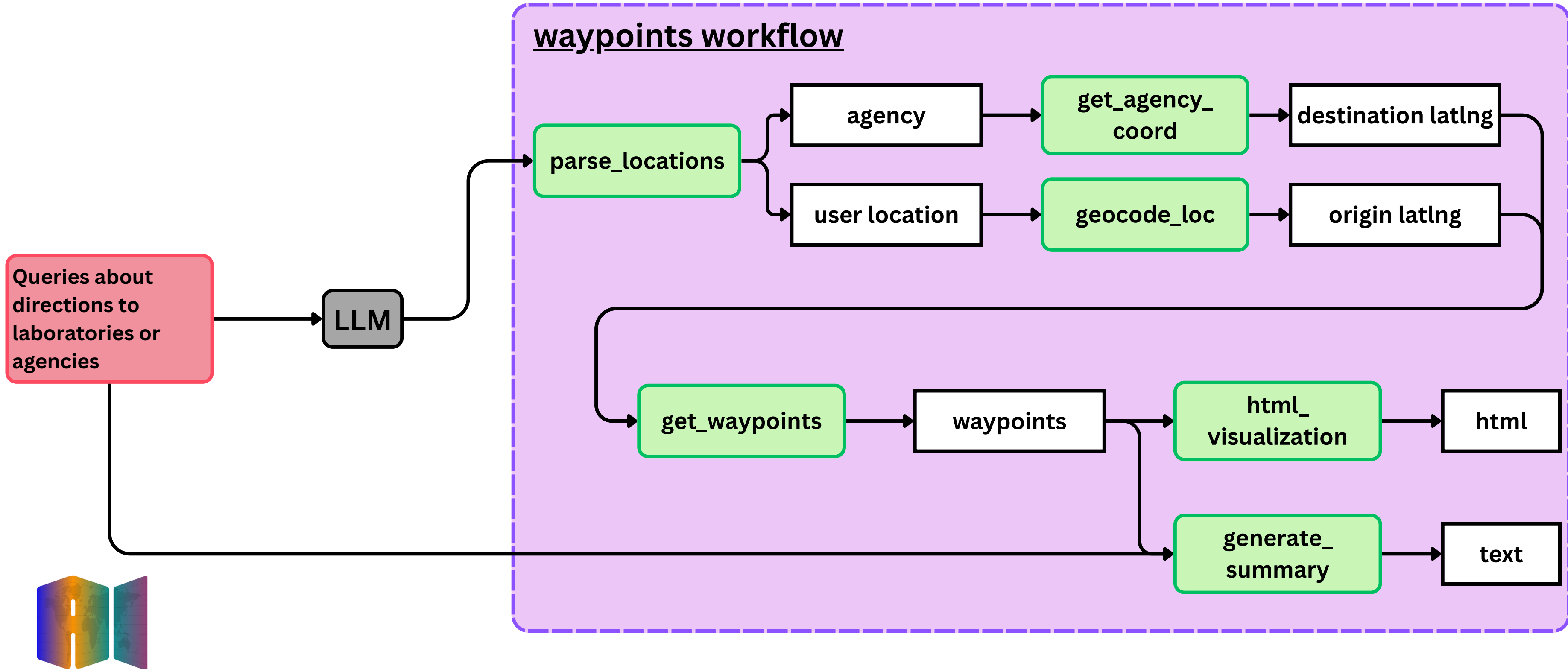
External API Workflow



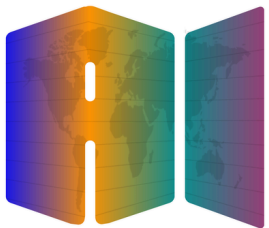
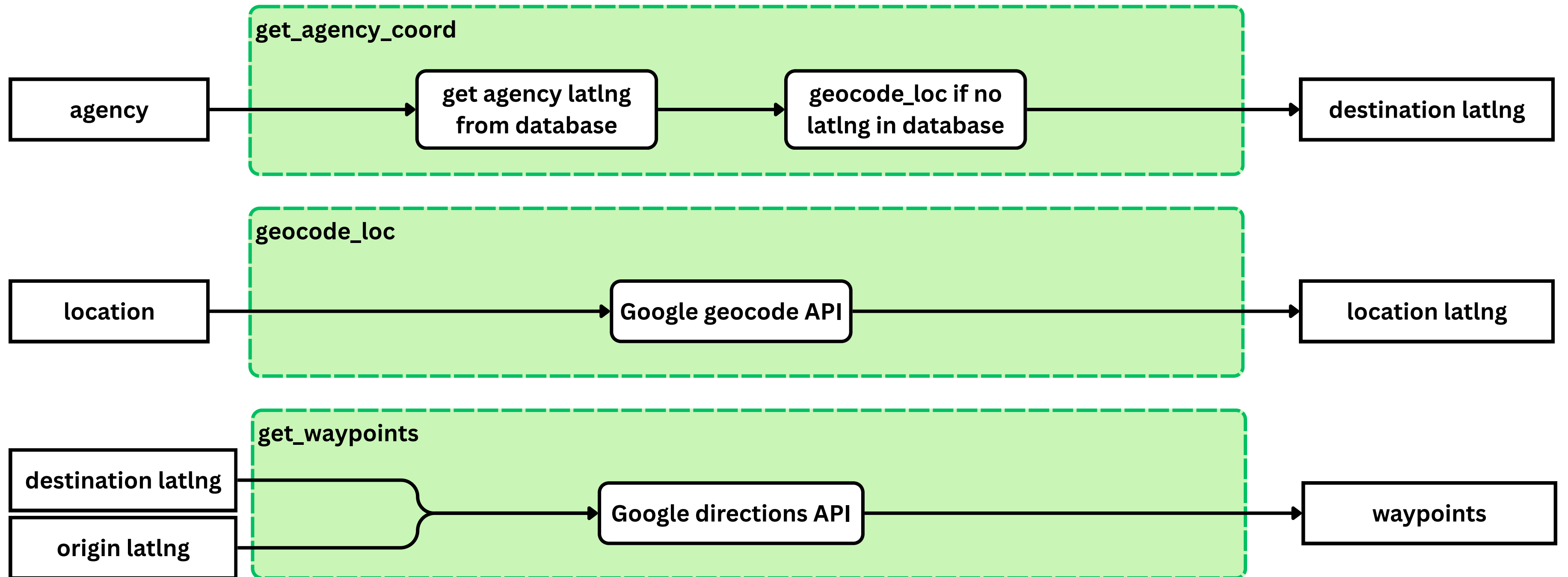
External API Workflow



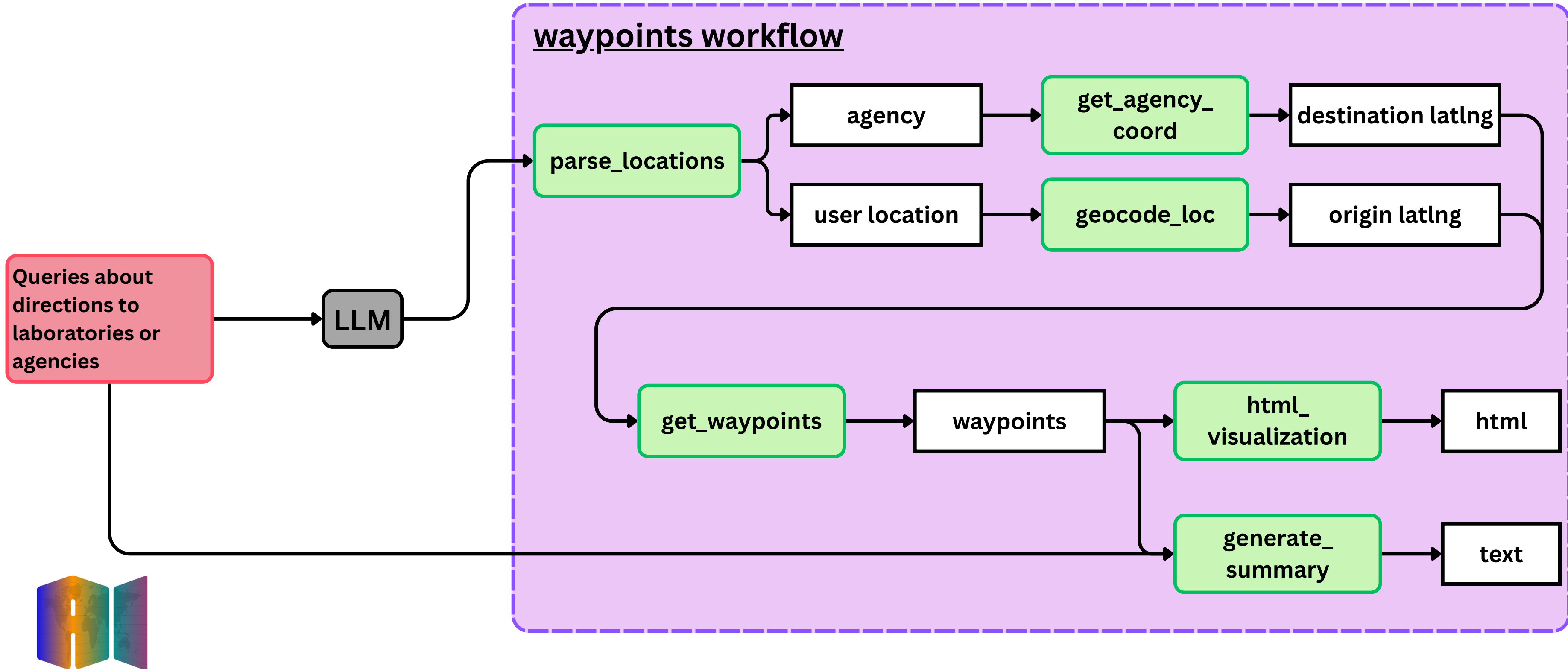
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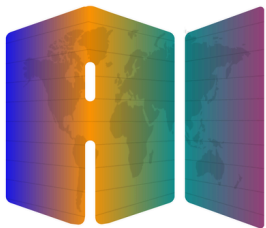
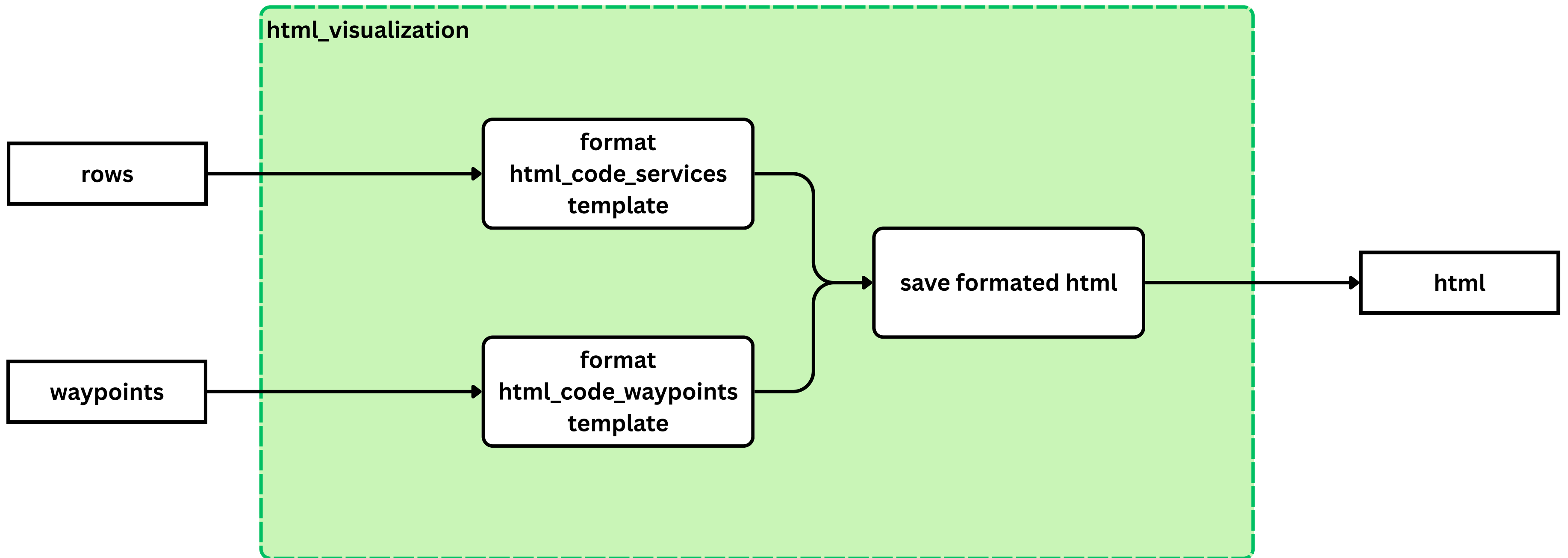
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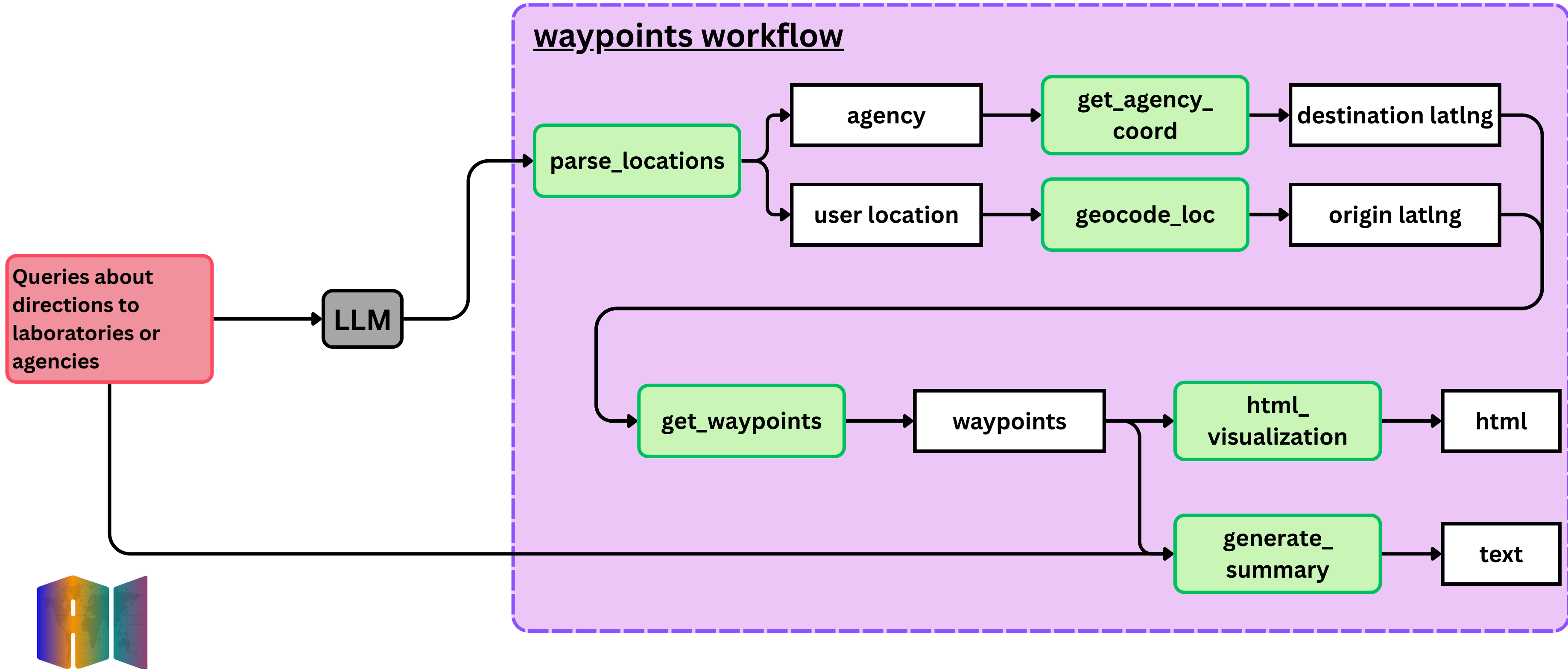
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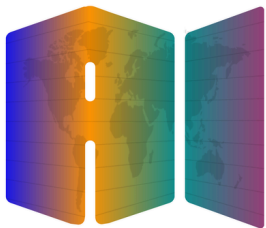
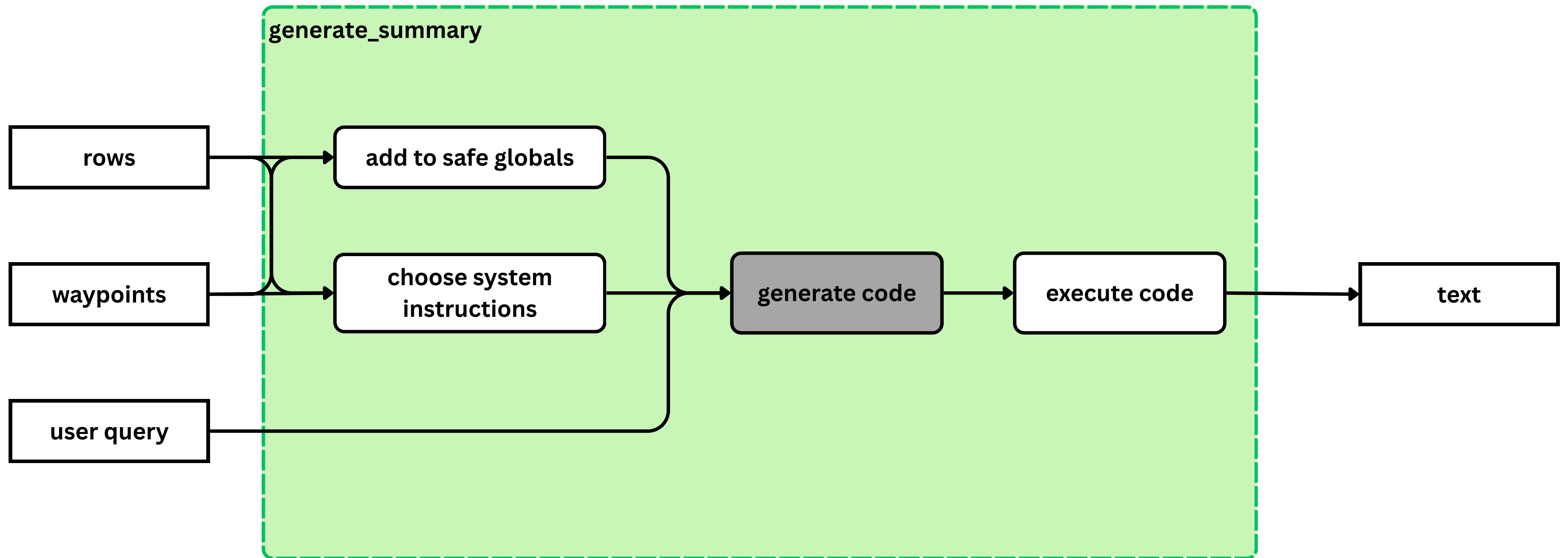
External API Workflow



External API Workflow

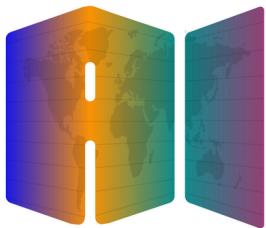


External API Workflow

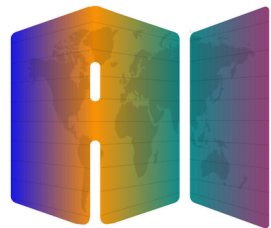
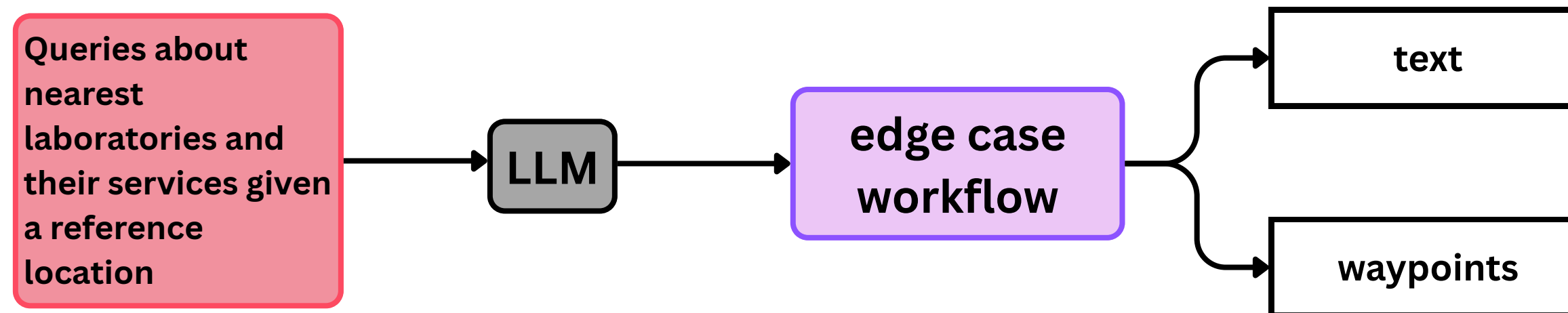


Workflow-as-a-tool Design Pattern

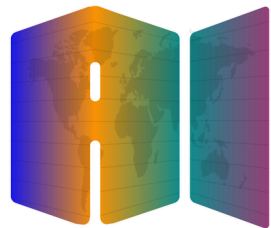
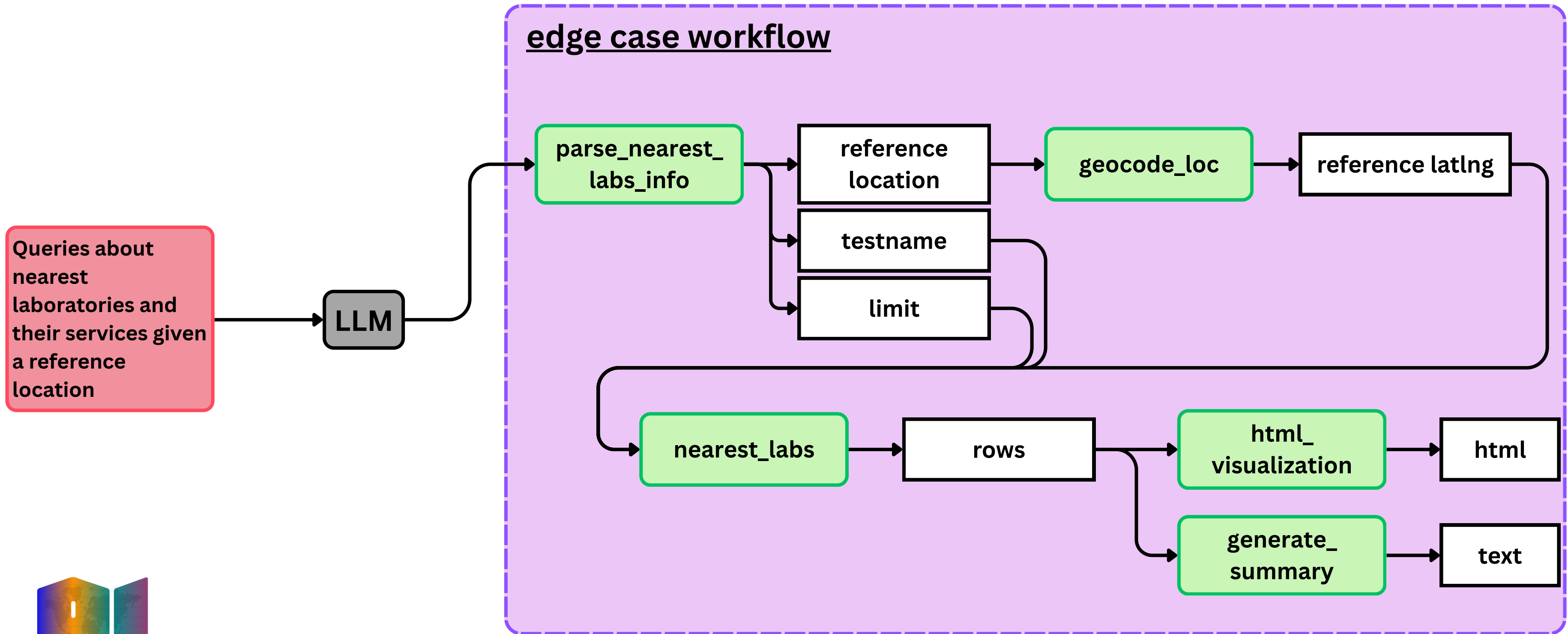
Edge Case Workflow



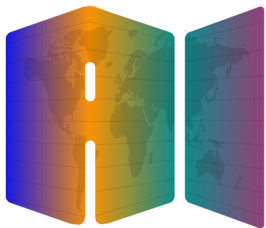
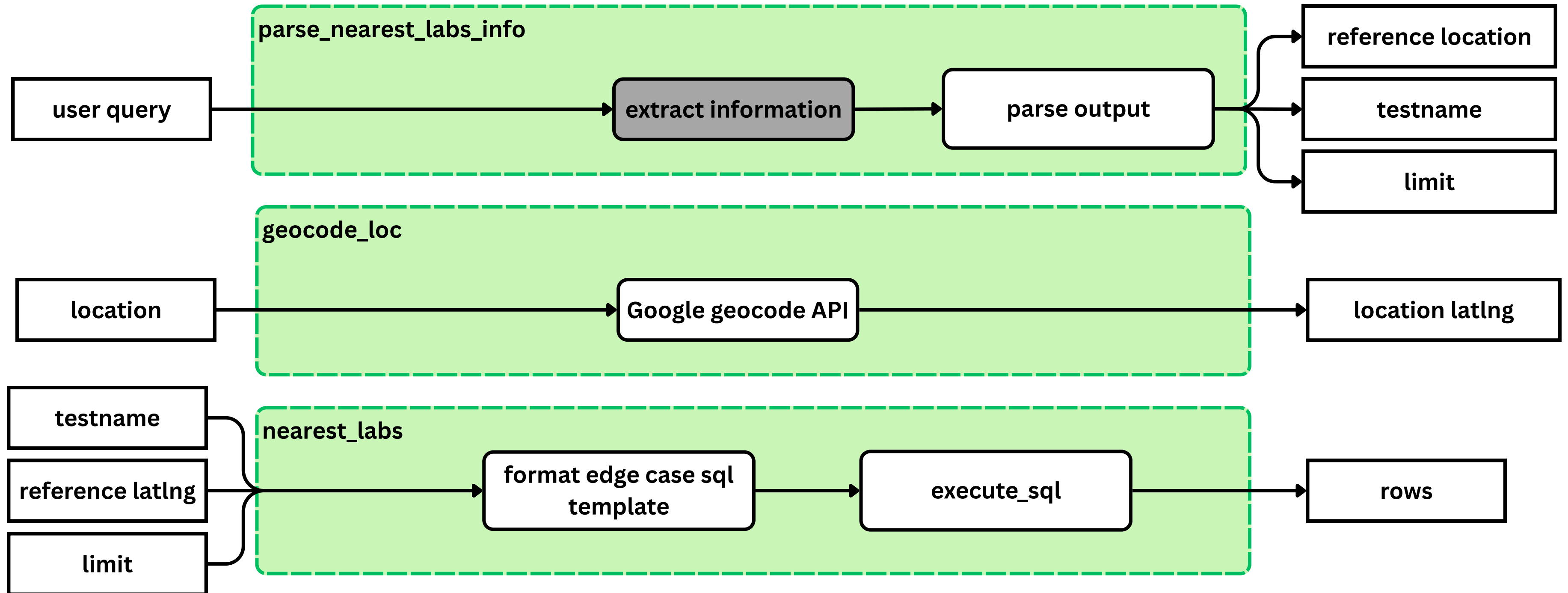
Edge Case Workflow



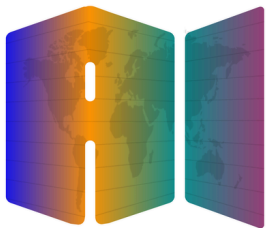
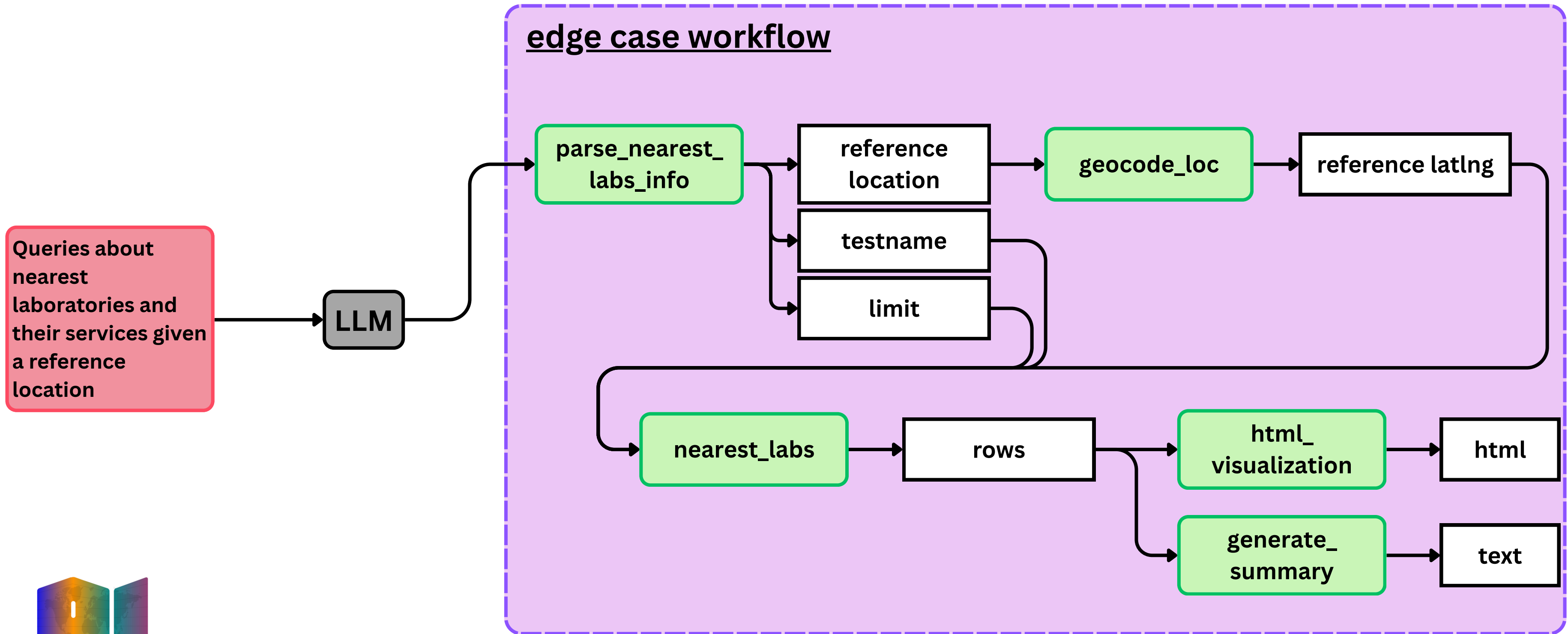
Edge Case Workflow



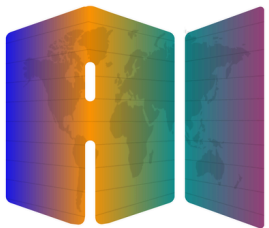
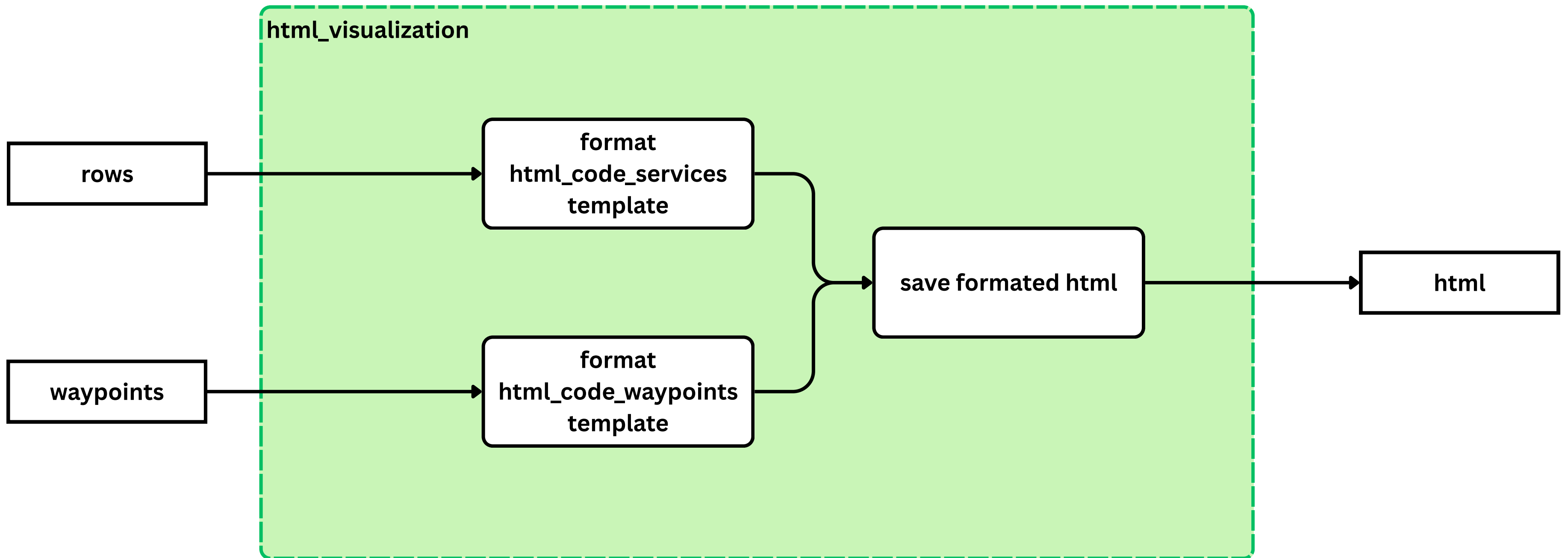
Edge Case Workflow



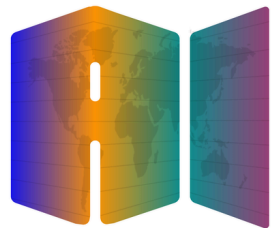
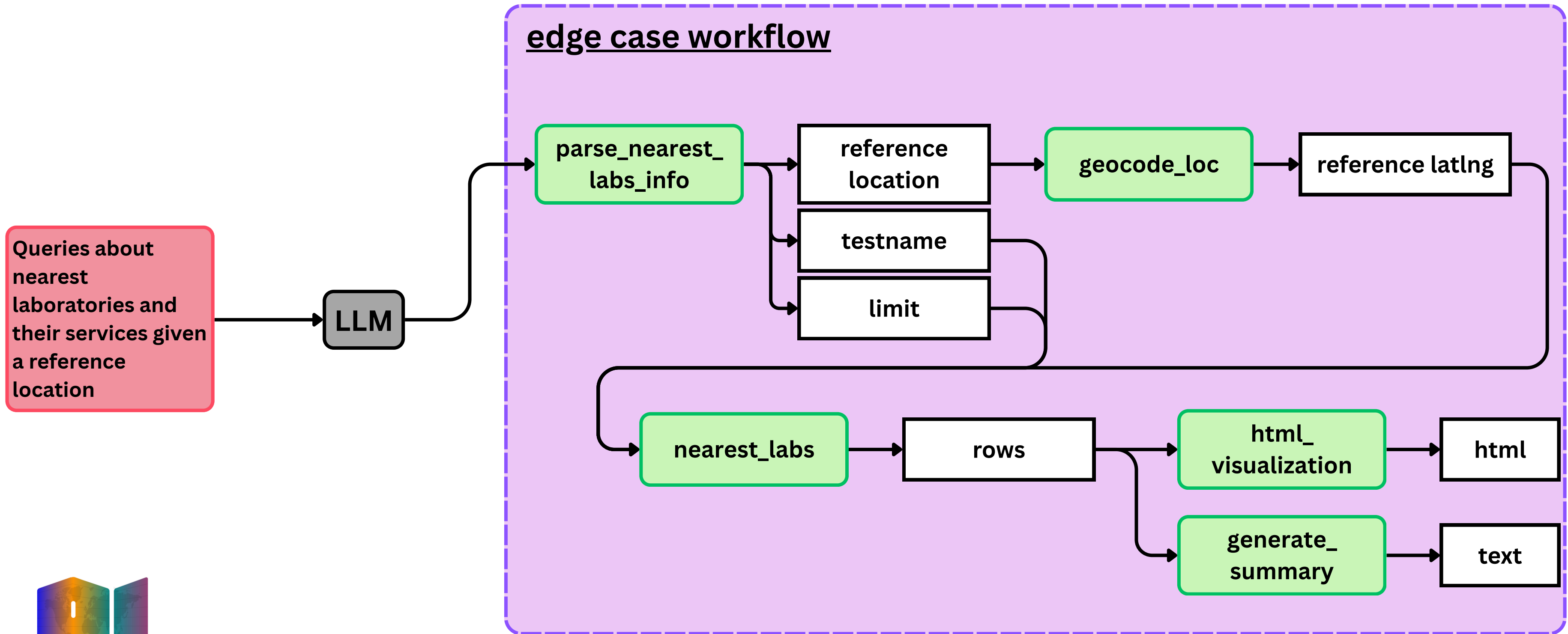
Edge Case Workflow



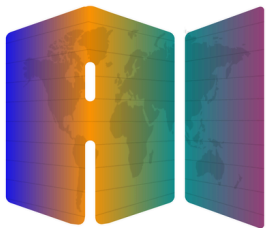
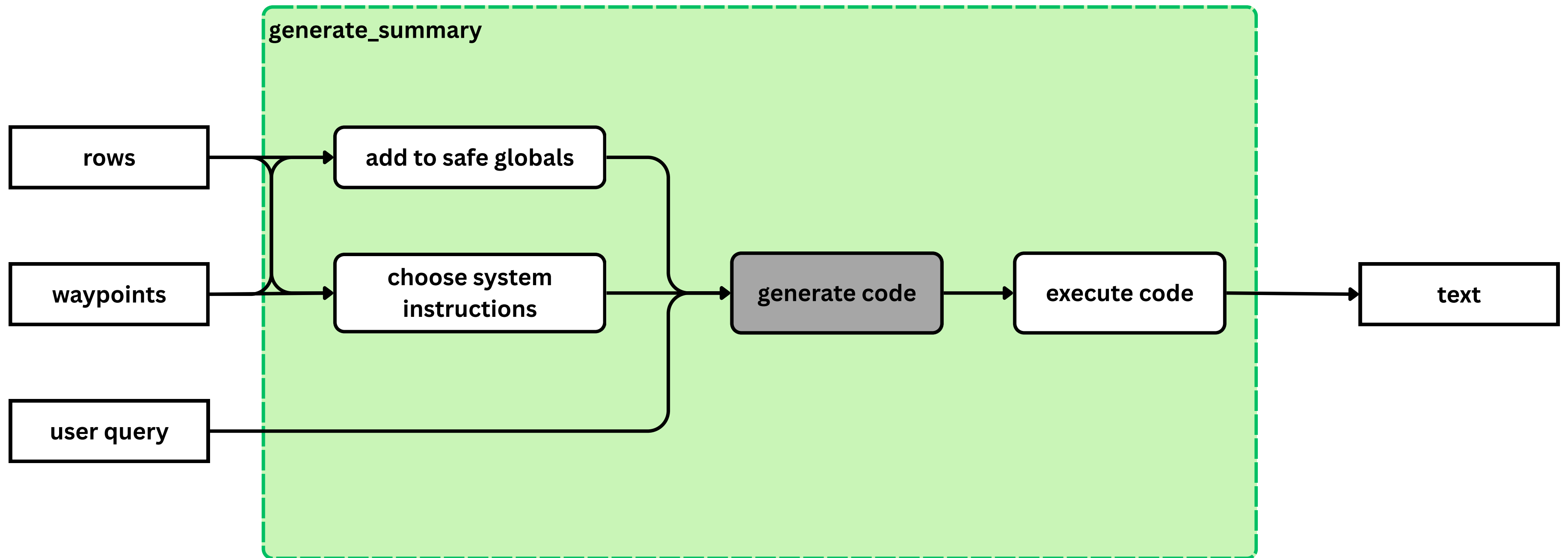
Edge Case Workflow



Edge Case Workflow

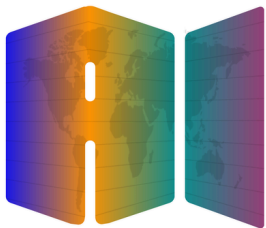


Edge Case Workflow

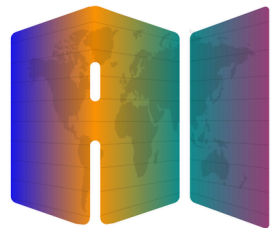
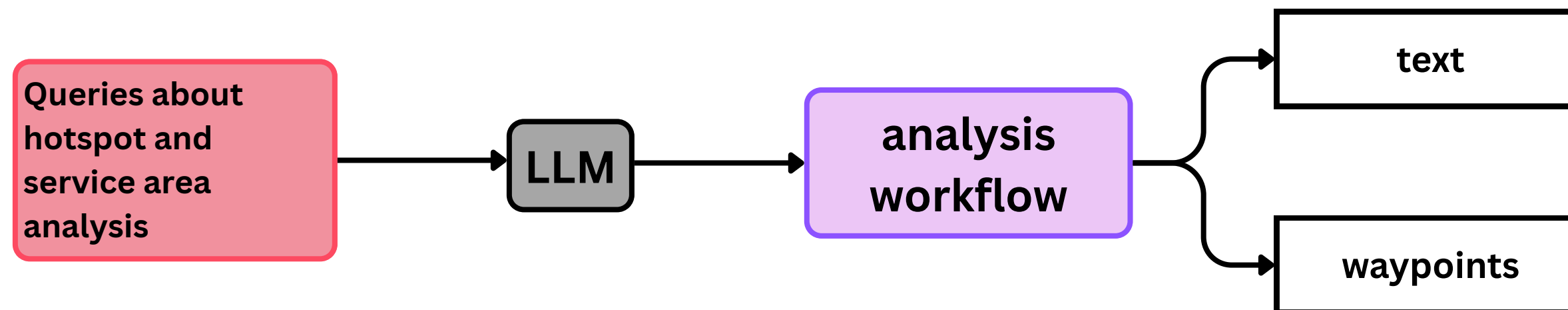


Workflow-as-a-tool Design Pattern

Analytics Workflow



Analysis Workflow



Analysis Workflow

DATA

OneLab Database Provincial Administrative Boundaries PSA Population data by province

HOTSPOT ANALYSIS

Simple Classification

No local OneLab presence	agency count is equal to zero
Potentially underserved	gap score is higher than at least 80% of all provinces
Laboratory/service hotspot	supply score is higher than at least 80% of all provinces
Well-served per capita	per capita supply score is higher than at least 80% of all provinces
Low coverage	supply score is lower than at least 80% of all provinces
Moderate coverage	fallback category

Simple Scores per province

values are normalized using minmax => $(\text{value} - \text{min_val}) / (\text{max_val} - \text{min_val})$

supply score- measures onelab presence

Supply Score = $\text{weight1} \times \text{normalized agency count} + \text{weight2} \times \text{normalized service count} + \text{weight3} \times \text{normalized unique test count}$

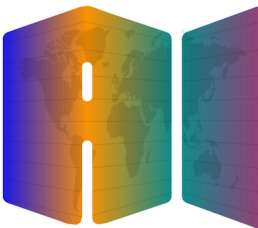
per capita supply score - measures supply relative to population

Per Capita Supply Score = $\text{weight4} \times \text{normalized labs per 100k} + \text{weight5} \times \text{normalized services per 100k} + \text{weight6} \times \text{normalized unique tests per 100k}$

gap score - uses population as a simple proxy for demand

Gap Score = $\text{normalized population} - \text{supply score}$

Higher gap score means the province may be underserved relative to its population



Analysis Workflow

DATA

OneLab Database

Provincial Administrative Boundaries

PSA Population data by province

SERVICE AREA ANALYSIS

Provincial Service Area Band

Local OneLab presence

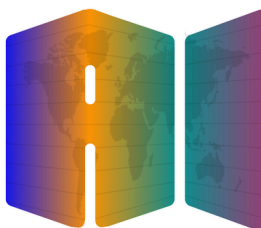
No local lab; nearest ≤ 50 km

No local lab; nearest 50-100 km

No local lab; nearest 100-200 km

No local lab; nearest > 200 km

distances were calculated using approximate straight line distance from province centroid



Analysis Workflow

This analysis data can identify:

1. Provinces with high laboratory concentration
2. provinces with high service availability
3. Provinces with broad test coverage
4. Provinces that appear underserved relative to population
5. Provinces without local onelab presence
6. Approximate province-level service area coverage
7. Provinces with at least one local onelab agency
8. Population living in provinces with local onelab presence
9. Provinces with no local onelab presence
10. Approximate distance from each province centroid to onelab agency
11. Distance band classification

Interpretation notes

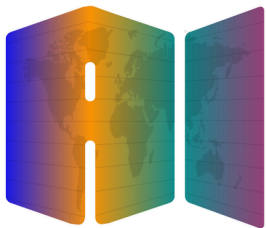
The results should be interpreted as a supply-side and population-normalized analysis.

The analysis can identify provinces that appear underserved relative to population, but it does not directly measure actual testing demand.

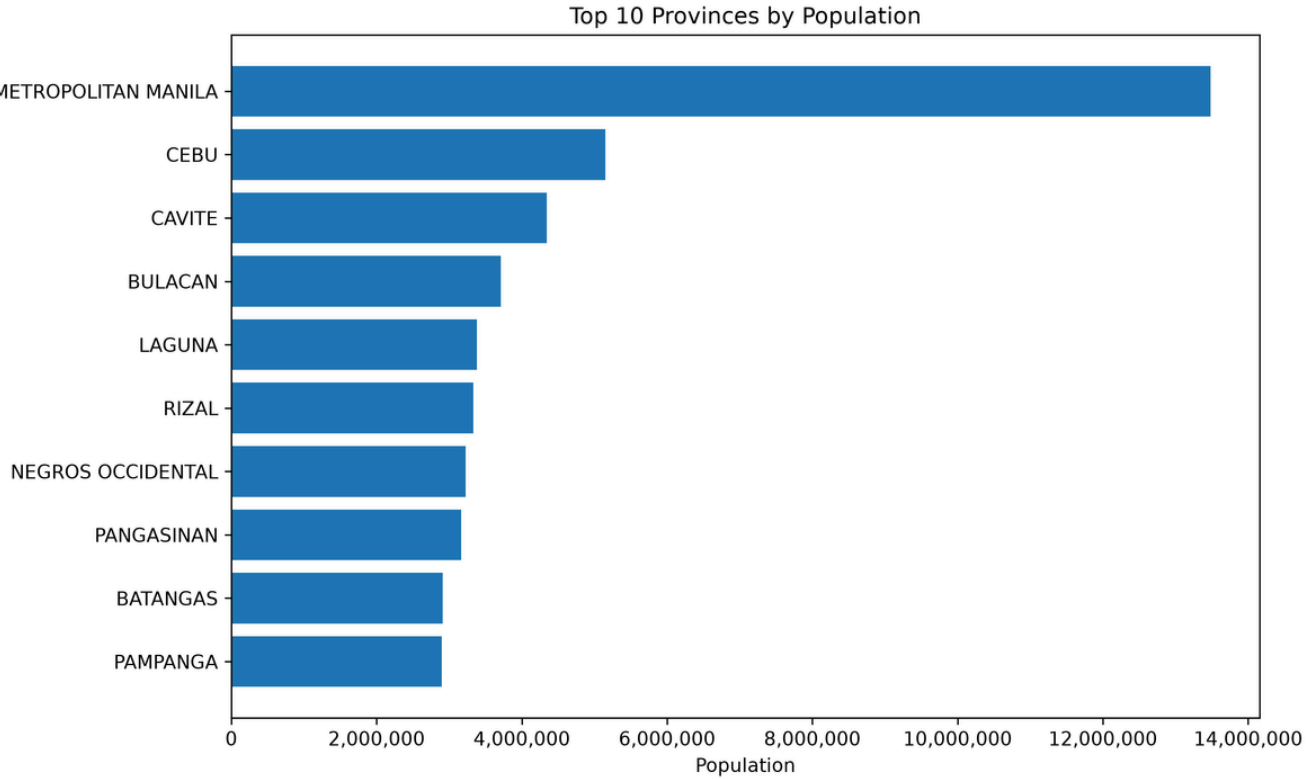
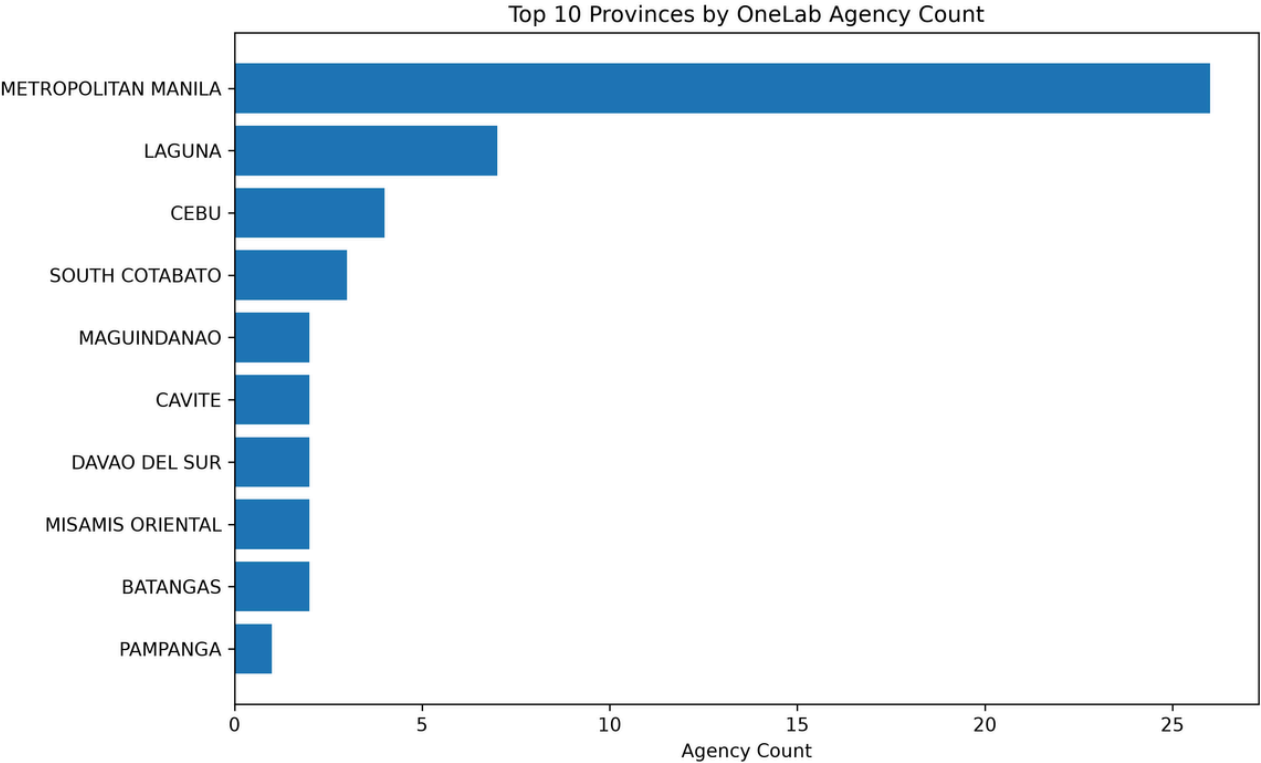
Therefore, provinces identified as potentially underserved should be treated as candidates for further validation, not final.

Limitations

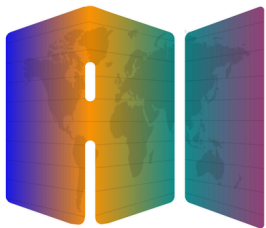
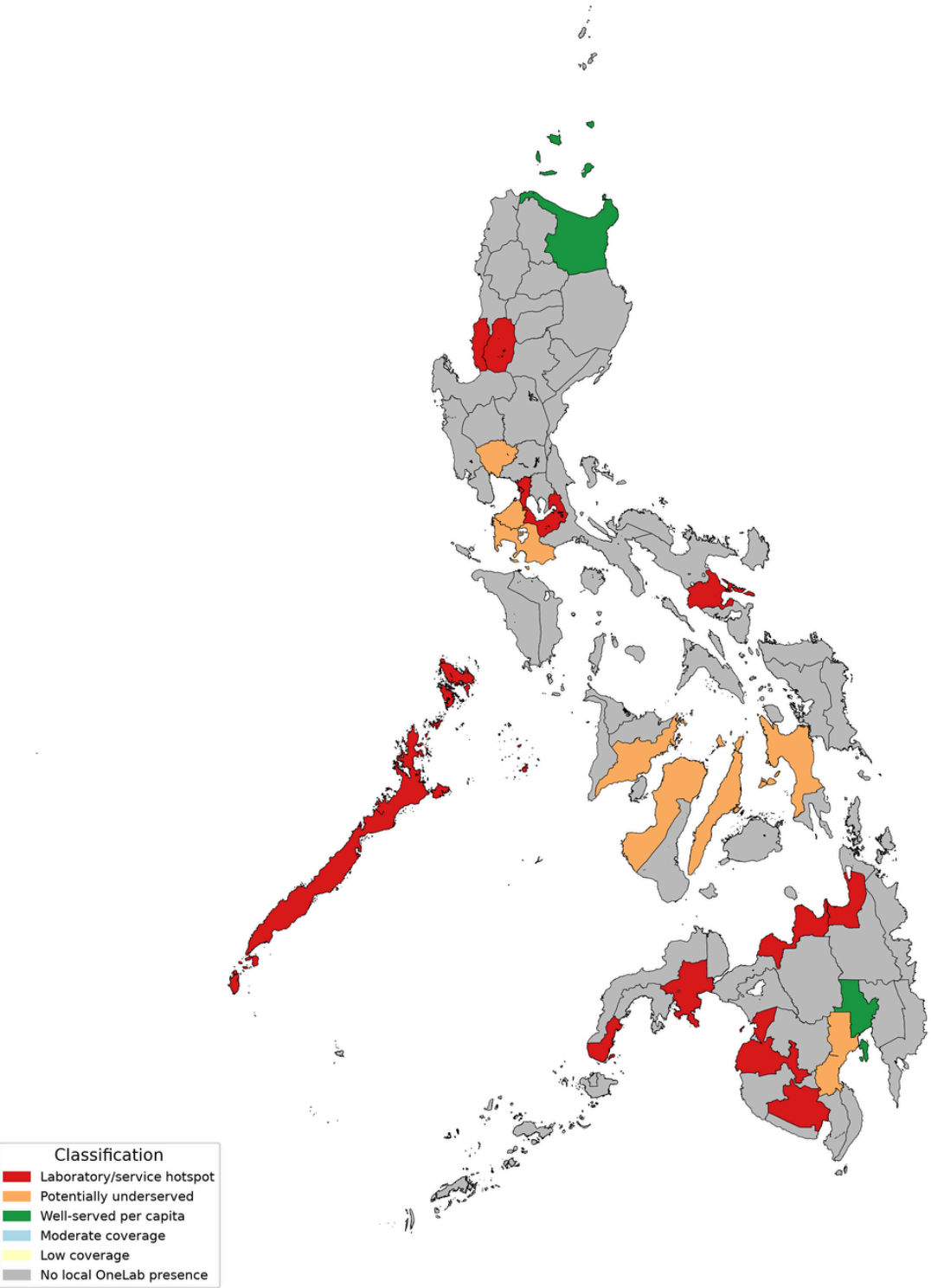
1. Population is used as a simple proxy for demand
2. Travel time is not calculated
3. Road networks are not included
4. The service area distance is based on approximate straight-line distance.
5. Province-level aggregation may hide city-level or municipal-level gaps.
6. The analysis assumes that agency coordinates are accurate.
7. The analysis does not account for laboratory capacity, accreditation, turnaround time, or actual workload
8. Fee differences may reflecty differences in test complexity, not simply affordability.



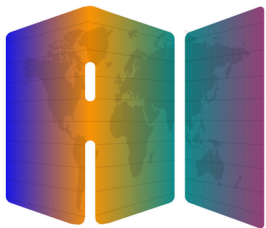
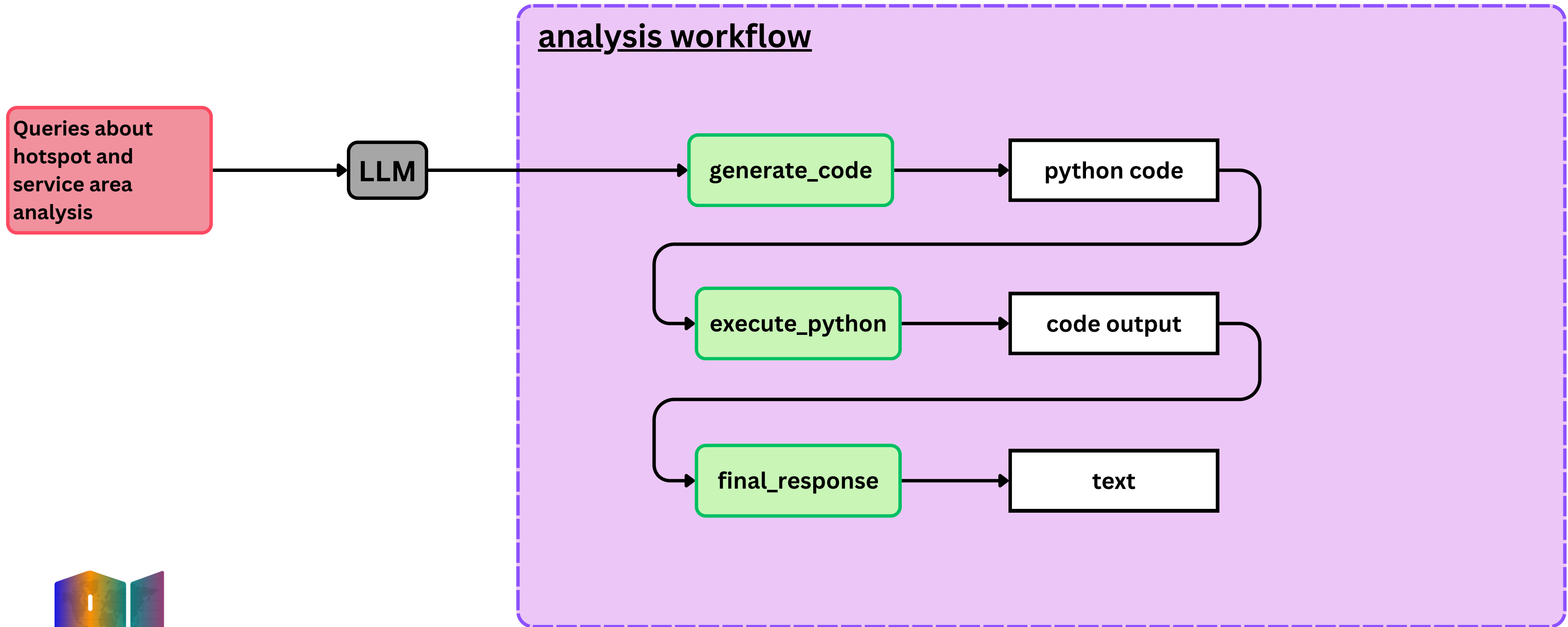
Analysis Workflow



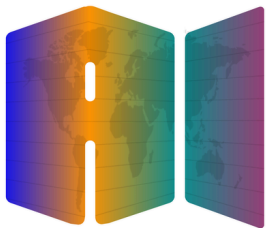
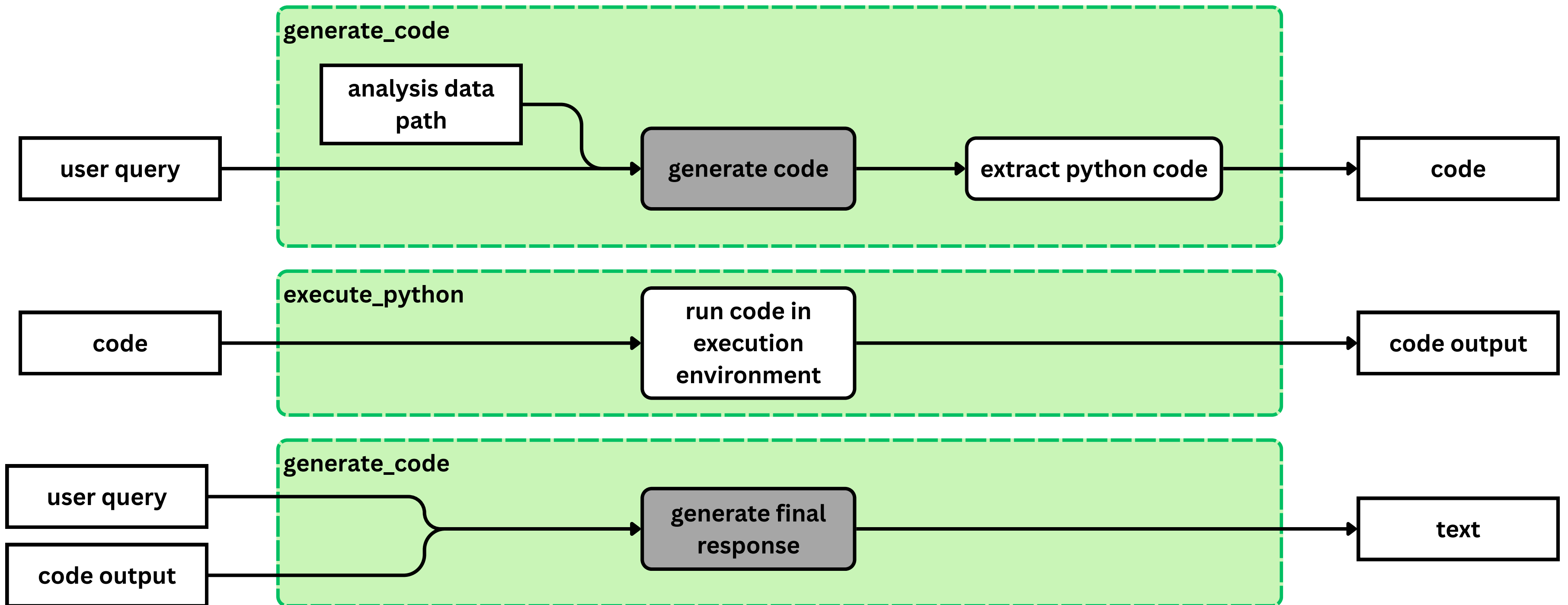
Province-Level OneLab Hotspot Classification



Analysis Workflow



Analysis Workflow



Workflow-as-a-tool Design Pattern

THE END

